

MSU: CHMY 362 ELEMENTS OF PHYSICAL CHEMISTRY



Jane Klassen

Montana State University in Bozeman

Montana State University in Bozeman
MSU: CHMY 362 Elements of Physical
Chemistry

Jane Klassen

This text is disseminated via the Open Education Resource (OER) LibreTexts Project (<https://LibreTexts.org>) and like the thousands of other texts available within this powerful platform, it is freely available for reading, printing, and "consuming."

The LibreTexts mission is to bring together students, faculty, and scholars in a collaborative effort to provide an accessible, and comprehensive platform that empowers our community to develop, curate, adapt, and adopt openly licensed resources and technologies; through these efforts we can reduce the financial burden born from traditional educational resource costs, ensuring education is more accessible for students and communities worldwide.

Most, but not all, pages in the library have licenses that may allow individuals to make changes, save, and print this book. Carefully consult the applicable license(s) before pursuing such effects. Instructors can adopt existing LibreTexts texts or Remix them to quickly build course-specific resources to meet the needs of their students. Unlike traditional textbooks, LibreTexts' web based origins allow powerful integration of advanced features and new technologies to support learning.



LibreTexts is the adaptable, user-friendly non-profit open education resource platform that educators trust for creating, customizing, and sharing accessible, interactive textbooks, adaptive homework, and ancillary materials. We collaborate with individuals and organizations to champion open education initiatives, support institutional publishing programs, drive curriculum development projects, and more.

The LibreTexts libraries are Powered by [NICE CXone Expert](#) and was supported by the Department of Education Open Textbook Pilot Project, the California Education Learning Lab, the UC Davis Office of the Provost, the UC Davis Library, the California State University Affordable Learning Solutions Program, and Merlot. This material is based upon work supported by the National Science Foundation under Grant No. 1246120, 1525057, and 1413739.

Any opinions, findings, and conclusions or recommendations expressed in this material are those of the author(s) and do not necessarily reflect the views of the National Science Foundation nor the US Department of Education.

Have questions or comments? For information about adoptions or adaptations contact info@LibreTexts.org or visit our main website at <https://LibreTexts.org>.

This text was compiled on 01/12/2026

TABLE OF CONTENTS

Licensing

1: Calorimetry

- 1.1: Calorimetry header
- 1.2: What is Calorimetry?
- 1.3: Constant Volume Calorimetry

2: Viscosity

- 2.1: What is Viscosity?
- 2.2: Viscosity Theory

3: Surface Tension

- 3.1: What is surface tension?
- 3.2: Cohesive and Adhesive Forces
- 3.3: Contact Angles
- 3.4: Capillary Action

4: MALDI-TOF

- 4.1: Overview of mass spectrometry
- 4.2: Determining the formula of a fragment based on the mass of the Peak
- 4.3: The M+1 Peak

5: Fluorometry

- 5.1: Fluorescence and Phosphorescence

6: UV-Vis

- 6.1: What Causes Molecules to Absorb UV and Visible Light
- 6.2: Solar Energy - UVA, UVB, UVC
- 6.3: Sunscreen

Index

Glossary

Detailed Licensing

Licensing

A detailed breakdown of this resource's licensing can be found in [Back Matter/Detailed Licensing](#).

CHAPTER OVERVIEW

1: Calorimetry

1.1: Calorimetry header

1.2: What is Calorimetry?

1.3: Constant Volume Calorimetry

1: Calorimetry is shared under a [not declared](#) license and was authored, remixed, and/or curated by LibreTexts.

SECTION OVERVIEW

1.1: Calorimetry header

Calorimetry is the process of measuring the amount of heat released or absorbed during a chemical reaction. By knowing the change in heat, it can be determined whether or not a reaction is exothermic (releases heat) or endothermic (absorbs heat). Calorimetry also plays a large part of everyday life, controlling the metabolic rates in humans and consequently maintaining such functions like body temperature.

1.1: Calorimetry header is shared under a [not declared](#) license and was authored, remixed, and/or curated by LibreTexts.

1.2: What is Calorimetry?

Calorimetry is the experimental science of measuring the heat changes that accompany chemical or physical changes. The accurate measurement of small amounts of heat is experimentally challenging. Nevertheless, calorimetry is an area in which great experimental sophistication has been achieved and remarkably accurate measurements can be made. Numerous devices have been developed to measure heat changes.

Some of these devices measure a (usually small) temperature change. Such devices are calibrated by measuring how much their temperature increases when a known amount of heat is introduced. This is usually accomplished by passing a known electric current through a known resistance for a known time. Other calorimeters measure the amount of some substance that undergoes a phase change. The ice calorimeter is an important example of the latter method. In an ice calorimeter, the heat of the process is transferred to a mixture of ice and water. The amount of ice that melts is a direct measure of the amount of heat released by the process. The amount of ice melted can be determined either by direct measurement of the increase in the amount of water present or by measuring the change in the volume of the ice–water mixture. (Since ice occupies a greater volume than the same mass of water, melting is accompanied by a decrease in the total volume occupied by the mixture of ice and water.)

The processes that can be investigated accurately using calorimetry are limited by two important considerations. One is that the process must go to completion within a relatively short time. No matter how carefully it is constructed, any calorimeter will exchange thermal energy with its environment at some rate. If this rate is not negligibly small compared to the rate at which the process evolves heat, the accuracy of the measurement is degraded. The second limitation is that the process must involve complete conversion of the system from a known initial state to a known final state. When the processes of interest are chemical reactions, these considerations mean that the reactions must be quantitative and fast.

Combustion reactions and catalytic hydrogenation reactions usually satisfy these requirements, and they are the most commonly investigated. However, even in these cases, there can be complications. For a compound containing only carbon, hydrogen, and oxygen, combustion using excess oxygen produces only carbon dioxide and water. For compounds containing heteroatoms like nitrogen, sulfur, or phosphorus, there may be more than one heteroatom-containing product. For example, combustion of an organosulfur compound might produce both sulfur dioxide and sulfur trioxide. To utilize the thermochemical data obtained in such an experiments, a chemical analysis must be done to determine the amount of each oxide present.

This page titled [1.2: What is Calorimetry?](#) is shared under a [CC BY-SA 4.0](#) license and was authored, remixed, and/or curated by [Paul Ellgen](#) via [source content](#) that was edited to the style and standards of the LibreTexts platform.

- **8.9: Calorimetry** by Paul Ellgen is licensed [CC BY-SA 4.0](#). Original source: <https://www.amazon.com/Thermodynamics-Chemical-Equilibrium-Paul-Ellgen/dp/1492114278>.

1.3: Constant Volume Calorimetry

Learning Objectives

Make sure you thoroughly understand the following essential concept:

- Describe a simple *calorimeter* and explain how it is employed and how its heat capacity is determined.
- Perform calculations necessary to determine heat content as determined by a calorimeter.

Constant Volume Calorimetry, also known as bomb calorimetry, is used to measure the heat of a reaction while holding volume constant and resisting large amounts of pressure. Although these two aspects of bomb calorimetry make for accurate results, they also contribute to the difficulty of bomb calorimetry. In this module, the basic assembly of a bomb calorimeter will be addressed, as well as how bomb calorimetry relates to the heat of reaction and heat capacity and the calculations involved in regards to these two topics.

Introduction

Calorimetry is used to measure quantities of heat, and can be used to determine the heat of a reaction through experiments. Usually a **coffee-cup calorimeter** is used since it is simpler than a bomb calorimeter, but to measure the heat evolved in a combustion reaction, constant volume or bomb calorimetry is ideal. A constant volume calorimeter is also more accurate than a coffee-cup calorimeter, but it is more difficult to use since it requires a well-built reaction container that is able to withstand large amounts of pressure changes that happen in many chemical reactions.

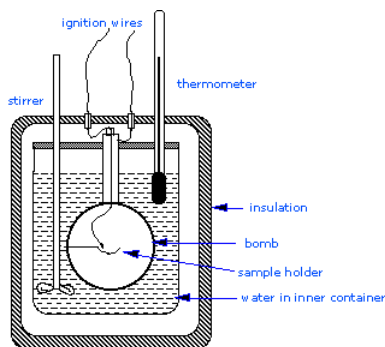


Figure 1.3.1: A simplified bomb calorimeter.

Most serious calorimetry carried out in research laboratories involves the determination of **heats of combustion** $\Delta H_{\text{combustion}}$, since these are essential to the determination of standard enthalpies of formation of the thousands of new compounds that are prepared and characterized each month. In a constant volume calorimeter, the system is sealed or isolated from its surroundings, which accounts for why its volume is fixed and there is no volume-pressure work done. A bomb calorimeter structure consists of the following:

- Steel bomb which contains the reactants
- Water bath in which the bomb is submerged
- Thermometer
- A motorized stirrer
- Wire for ignition

Since the process takes place at constant volume, the reaction vessel must be constructed to withstand the high pressure resulting from the combustion process, which amounts to a confined explosion. The vessel is usually called a “bomb”, and the technique is known as *bomb calorimetry*. The reaction is initiated by discharging a capacitor through a thin wire which ignites the mixture.

Another consequence of the constant-volume condition is that the heat released corresponds to q_v , and thus to the internal energy change ΔU rather than to ΔH . The enthalpy change is calculated according to the formula

$$\Delta H = q_v + \Delta n_g RT \quad (1.3.1)$$

where Δn_g is the change in the number of moles of gases in the reaction.

✓ Example 1.3.1: Combustion of Biphenyl

A sample of biphenyl ($(C_6H_5)_2$) weighing 0.526 g was ignited in a bomb calorimeter initially at 25°C, producing a temperature rise of 1.91 K. In a separate calibration experiment, a sample of benzoic acid (C_6H_5COOH) weighing 0.825 g was ignited under identical conditions and produced a temperature rise of 1.94 K. For benzoic acid, the heat of combustion at constant volume is known to be 3,226 kJ mol⁻¹ (that is, $\Delta U = -3,226$ kJ mol⁻¹.) Use this information to determine the standard enthalpy of combustion of biphenyl.

Solution

Begin by working out the calorimeter constant (heat capacity of the bomb):

- Moles of benzoic acid:

$$\frac{0.825 \text{ g}}{122.1 \text{ g/mol}} = 0.00676 \text{ mol}$$

- Heat released to calorimeter:

$$\begin{aligned} \text{moles benzoic acid} \times \Delta H_{\text{combustion}} &= \Delta U_{\text{combustion}} \\ (0.00676 \text{ mol}) \times (3226 \text{ kJ/mol}) &= 21.80 \text{ kJ} \end{aligned}$$

- Calorimeter constant (W):

$$\frac{21.80 \text{ kJ}}{1.94 \text{ K}} = 11.24 \text{ kJ/K}$$

Now determine $\Delta U_{\text{combustion}}$ of the biphenyl ("BP"):

- moles of biphenyl:

$$\frac{0.526 \text{ g}}{154.12 \text{ g/mol}} = 0.00341 \text{ mol}$$

- heat released to calorimeter:

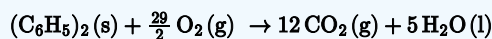
$$\begin{aligned} \Delta T \times W &= q_v \\ (1.91 \text{ K}) \times (11.24 \text{ kJ/K}) &= 21.46 \text{ kJ} \end{aligned} \tag{1.3.2}$$

- heat released per mole of biphenyl:

$$\begin{aligned} \frac{21.46 \text{ kJ}}{0.00341 \text{ mol}} &= 6,293 \text{ kJ/mol} \\ \Delta U_{\text{combustion}}(\text{BP}) &= -6,293 \text{ kJ/mol} \end{aligned}$$

This is the heat change at constant volume, q_v ; the negative sign indicates that the reaction is exothermic, as all combustion reactions are.

From the balanced reaction equation



we can calculate the change in the moles of gasses for this reaction

$$\Delta n_g = 12 - \frac{29}{2} = -\frac{5}{2}$$

Thus the volume of the system *decreases* when the reaction takes place. Converting to ΔH , we can write the following equation. Additionally, recall that at constant volume, $\Delta U = q_v$.

$$\begin{aligned}
 \Delta H &= q_v + \Delta n_g RT \\
 &= \Delta U - \left(\frac{5}{2}\right) (8.314 \text{ J mol}^{-1} \text{ K}^{-1})(298 \text{ K}) \\
 &= (-6,293 \text{ kJ/mol}) - (6,194 \text{ J/mol}) \\
 &= (-6,293 - 6.2) \text{ kJ/mol} \\
 &= -6299 \text{ kJ/mol}
 \end{aligned}$$

A common mistake here is to forget that the subtracted term is in J, not kJ. Note that the additional 6.2 kJ in ΔH compared to ΔU reflects the work that the surroundings do on the system as the volume of gases decreases according to the reaction equation.

Determining the Heat of Reaction

The amount of heat that the system gives up to its surroundings so that it can return to its initial temperature is the **heat of reaction**. The heat of reaction is just the negative of the thermal energy gained by the calorimeter and its contents ($q_{\text{calorimeter}}$) through the combustion reaction.

$$q_{\text{rxn}} = -q_{\text{calorimeter}} \quad (1.3.3)$$

where

$$q_{\text{calorimeter}} = q_{\text{bomb}} + q_{\text{water}} \quad (1.3.4)$$

If the constant volume calorimeter is set up the same way as before, (same steel bomb, same amount of water, etc.) then the heat capacity of the calorimeter can be measured using the following formula:

$$q_{\text{calorimeter}} = (\text{heat capacity of calorimeter}) \times \Delta T \quad (1.3.5)$$

Heat capacity is defined as the amount of heat needed to increase the temperature of the entire calorimeter by 1 °C. The equation above can also be used to calculate q_{rxn} from $q_{\text{calorimeter}}$ calculated by Equation 1.3.3. The heat capacity of the calorimeter can be determined by conducting an experiment.

✓ Example 1.3.4: Heat of Combustion

1.150 g of sucrose goes through combustion in a bomb calorimeter. If the temperature rose from 23.42 °C to 27.64 °C and the heat capacity of the calorimeter is 4.90 kJ/°C, then determine the heat of combustion of sucrose, $\text{C}_{12}\text{H}_{22}\text{O}_{11}$ (in kJ per mole of $\text{C}_{12}\text{H}_{22}\text{O}_{11}$).

Solution

Given:

- mass of $\text{C}_{12}\text{H}_{22}\text{O}_{11}$: 1.150 g
- T_{initial} : 23.42°C
- T_{final} : 27.64°C
- Heat Capacity of Calorimeter: 4.90 kJ/°C

Using Equation 1.3.5 to calculate $q_{\text{calorimeter}}$:

$$\begin{aligned}
 q_{\text{calorimeter}} &= (4.90 \text{ kJ/}^\circ\text{C}) \times (27.64 - 23.42)^\circ\text{C} \\
 &= (4.90 \times 4.22) \text{ kJ} = 20.7 \text{ kJ}
 \end{aligned}$$

Plug into Equation 1.3.3:

$$\begin{aligned}
 q_{\text{rxn}} &= -q_{\text{calorimeter}} \\
 &= -20.7 \text{ kJ}
 \end{aligned}$$

But the question asks for kJ/mol $\text{C}_{12}\text{H}_{22}\text{O}_{11}$, so this needs to be converted:

$$q_{rxn} = \frac{-20.7 \text{ kJ}}{1.150 \text{ g C}_{12}\text{H}_{22}\text{O}_{11}}$$

$$= \frac{-18.0 \text{ kJ}}{\text{g C}_{12}\text{H}_{22}\text{O}_{11}}$$

Convert to per Mole $\text{C}_{12}\text{H}_{22}\text{O}_{11}$:

$$q_{rxn} = \frac{-18.0 \text{ kJ}}{\text{g C}_{12}\text{H}_{22}\text{O}_{11}} \times \frac{342.3 \text{ g C}_{12}\text{H}_{22}\text{O}_{11}}{1 \text{ mol C}_{12}\text{H}_{22}\text{O}_{11}}$$

$$= \frac{-6.16 \times 10^3 \text{ kJ}}{\text{mol C}_{12}\text{H}_{22}\text{O}_{11}}$$

Although calorimetry is simple in principle, its practice is a highly exacting art, especially when applied to processes that take place slowly or involve very small heat changes, such as the germination of seeds. Calorimeters can be as simple as a foam plastic coffee cup, which is often used in student laboratories. Research-grade calorimeters, able to detect minute temperature changes, are more likely to occupy table tops, or even entire rooms.

This page titled [1.3: Constant Volume Calorimetry](#) is shared under a [CC BY 3.0](#) license and was authored, remixed, and/or curated by [Stephen Lower](#) via [source content](#) that was edited to the style and standards of the LibreTexts platform.

- **14.5: Calorimetry** by [Stephen Lower](#) is licensed [CC BY 3.0](#). Original source: <http://www.chem1.com/acad/webtext/virtualtextbook.html>.

CHAPTER OVERVIEW

2: Viscosity

[2.1: What is Viscosity?](#)

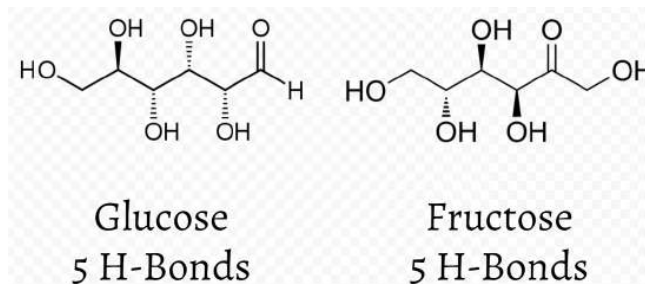
[2.2: Viscosity Theory](#)

2: Viscosity is shared under a [not declared](#) license and was authored, remixed, and/or curated by LibreTexts.

2.1: What is Viscosity?

Because its molecules can slide around each other, a liquid has the ability to flow. The resistance to such flow is called the **viscosity**. Liquids which flow very slowly, like glycerin or honey, have high viscosities. Those like ether or gasoline which flow very readily have low viscosities.

Viscosity is governed by the strength of intermolecular forces and especially by the shapes of the molecules of a liquid. Liquids whose molecules are polar or can form hydrogen bonds are usually more viscous than similar nonpolar substances. Honey, mostly glucose and fructose (see image below) is a good example of a liquid which owes its viscosity to hydrogen bonding.



Liquids containing long molecules are invariably very viscous. This is because the molecular chains get tangled up in each other like spaghetti—in order for the liquid to flow, the molecules must first unravel. Fuel oil, lubricating grease, and other long-chain alkane molecules are quite viscous for this reason. Glycerol, $\text{CH}_2\text{OHCHOHCH}_2\text{OH}$, is viscous partly because of the length of the chain but also because of the extensive possibilities for hydrogen bonding between the molecules. The video below shows several different long chained oils, each progressively more viscous.



The viscosity of a liquid always decreases as temperature increases. As the molecules acquire more energy, they can escape from their mutual traction more readily. Long-chain molecules can also wriggle around more freely at a higher temperature and hence disentangle more quickly. Below is a video that demonstrates this effect with a household liquid: honey. As a warning, the video has **loud** background music.



This page titled [2.1: What is Viscosity?](#) is shared under a [CC BY-NC-SA 4.0](#) license and was authored, remixed, and/or curated by [Ed Vitz, John W. Moore, Justin Shorb, Xavier Prat-Resina, Tim Wendorff, & Adam Hahn](#).

- [10.7: Viscosity](#) by Ed Vitz, John W. Moore, Justin Shorb, Xavier Prat-Resina, Tim Wendorff, & Adam Hahn is licensed [CC BY-NC-SA 4.0](#).

2.2: Viscosity Theory

Introduction

All liquids have a natural internal resistance to flow termed viscosity. Viscosity is the result of frictional interactions within a given liquid and is commonly expressed in two different ways.

Dynamic Viscosity

The first is dynamic viscosity, also known as absolute viscosity, which measures a fluid's resistance to flow. In precise terms, dynamic viscosity is the tangential force per unit area necessary to move one plane past another at unit velocity at unit distance apart. As one plane moves past another in a fluid, a velocity gradient is established between the two layers (Figure 2.2.1). Viscosity can be thought of as a drag coefficient proportional to this gradient.

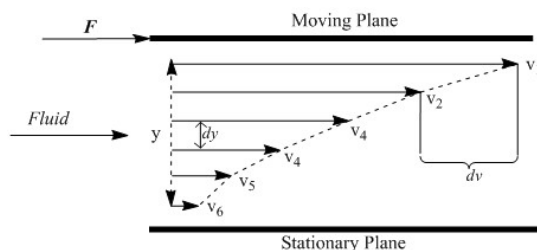


Figure 2.2.1 Fluid dynamics as one plane moves relative to a stationary plane through a liquid. The moving plane has area A and requires force F to overcome the fluid's internal resistance.

The force necessary to move a plane of area A past another in a fluid is given by Equation 2.2.1 where V is the velocity of the liquid, Y is the separation between planes, and η is the dynamic viscosity.

$$F = \eta A \frac{V}{Y} \quad (2.2.1)$$

V/Y also represents the velocity gradient (sometimes referred to as shear rate). Force over area is equal to τ , the shear stress, so the equation simplifies to Equation 2.2.2.

$$\tau = \eta \frac{V}{Y} \quad (2.2.2)$$

For situations where V does not vary linearly with the separation between plates, the differential formula based on Newton's equations is given in Equation 2.2.3.

$$\tau = \eta \frac{\delta V}{\delta Y} \quad (2.2.3)$$

Kinematic Viscosity

Kinematic viscosity, the other type of viscosity, requires knowledge of the density, ρ , and is given by Equation 2.2.4, where ν is the kinematic viscosity and the η is the dynamic viscosity.

$$\nu = \frac{\eta}{\rho} \quad (2.2.4)$$

Units of Viscosity

Viscosity is commonly expressed in Stokes, Poise, and SI units.

Dynamic Viscosity

The SI units for dynamic (absolute) viscosity is given in units of $\text{N}\cdot\text{s}/\text{m}^2$, $\text{Pa}\cdot\text{s}$, or $\text{kg}/(\text{m}\cdot\text{s})$, where N stands for Newton and Pa for Pascal. Poise are metric units expressed as $\text{dyne}\cdot\text{s}/\text{cm}^2$ or $\text{g}/(\text{m}\cdot\text{s})$. They are related to the SI unit by $\text{g}/(\text{m}\cdot\text{s}) = 1/10 \text{ Pa}\cdot\text{s}$. 100 centipoise, the centipoise (cP) being the most used unit of viscosity, is equal to one Poise. Table 2.2.1 shows the interconversion factors for dynamic viscosity.

Table 2.2.1: The interconversion factors for dynamic viscosity.

Table 2.2.2 lists the dynamic viscosities of several liquids at various temperatures in centipoise. The effect of the temperature on viscosity is clearly evidenced in the drastic drop in viscosity of water as the temperature is increased from near ambient to 60 degrees Celsius. Ketchup has a viscosity of 1000 cP at 30 degrees Celsius or more than 1000 times that of water at the same temperature!

Unit	Pa·s	Dyne·s/cm ² or g/(m·s) (Poise)	Centipoise (cP)
Pa·s	1	10	1000
Dyne·s/cm ² or g/(m·s) (Poise)	0.1	1	100
Centipoise (cP)	0.001	0.01	1

Table 2.2.2: Viscosities of common liquids (*at 0% evaporation volume).

Liquid	η (cP)	Temperature (°C)
Water	0.89	25
Water	0.47	60
Milk	2.0	18
Olive Oil	107.5	20
Toothpaste	70,000 - 100,000	18
Ketchup	1000	30
Custard	1,500	85-90
Crude Oil (WTI)*	7	15

Kinematic Viscosity

The CGS unit for kinematic viscosity is the Stoke which is equal to $10^{-4} \text{ m}^2/\text{s}$. Dividing by 100 yields the more commonly used centistoke. The SI unit for viscosity is m^2/s .

Newtonian versus Non-Newtonian Fluids

One of the invaluable applications of the determination of viscosity is identifying a given liquid as Newtonian or non-Newtonian in nature.

- Newtonian liquids are those whose viscosities remain constant for all values of applied shear stress.
- *Non-Newtonian* liquids are those liquids whose viscosities vary with applied shear stress and/or time.

Moreover, non-Newtonian liquids can be further subdivided into classes by their viscous behavior with shear stress:

- Pseudoplastic fluids whose viscosity decreases with increasing shear rate
- Dilatants in which the viscosity increases with shear rate.
- Bingham plastic fluids, which require some force threshold be surpassed to begin to flow and which thereafter flow proportionally to increasing shear stress.

Measuring Viscosity

Viscometers are used to measure viscosity. There are seven different classes of viscometer:

1. Capillary viscometers (available for CHMY 362).
2. Orifice viscometers (available for CHMY 362).
3. High temperature high shear rate viscometers.
4. Rotational viscometers.
5. Falling ball viscometers.
6. Vibrational viscometers.

7. Ultrasonic Viscometers.

Capillary Viscometers

Capillary viscometers are the most widely used viscometers when working with Newtonian fluids and measure the flow rate through a narrow, usually glass tube. In some capillary viscometers, an external force is required to move the liquid through the capillary; in this case, the pressure difference across the length of the capillary is used to obtain the viscosity coefficient.

Capillary viscometers require a liquid reservoir, a capillary of known dimensions, a pressure controller, a flow meter, and a thermostat be present. These viscometers include, Modified Ostwald viscometers, Suspended-level viscometers, and Reverse-flow viscometers and measure *kinematic* viscosity.

The equation governing this type of viscometry is the Poiseuille law (Equation 2.2.5), where Q is the overall flowrate, ΔP , the pressure difference, a , the internal radius of the tube, η , the dynamic viscosity, and l the path length of the fluid.

$$Q = \frac{\pi \Delta P a^4}{8 \eta l} \quad (2.2.5)$$

Here, Q is equal to V/t ; the volume of the liquid measured over the course of the experiment divided by the time required for it to move through the capillary where V is volume and t is time.

For gravity-type capillary viscometers, those relying on gravity to move the liquid through the tube rather than an applied force, Equation 2.2.6 is used to find viscosity, obtained by substituting the relation Equation 2.2.5 with the experimental values, where P is pressure, ρ is density, g is the gravitational constant, and h is the height of the column.

$$\eta = \frac{\pi g h a^4}{8 l V} \rho t \quad (2.2.6)$$

An example of a capillary viscometer (Ostwald viscometer) is shown in Figure 2.2.2.

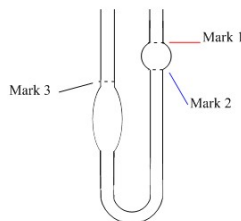


Figure 2.2.2 The capillary, submerged in an isothermal bath, is filled until the liquid lies at Mark 3. The liquid is then drawn up through the opposite side of the tube. The time it takes for the liquid to travel from Mark 2 to Mark 1 is used to compute the viscosity.

For Ostwald viscometers used in CHMY 362, the viscometer can be calibrated by using a liquid with known density, ρ , and viscosity, η , (e.g., water). Equation 2.2.6 is then simplified to Equation 2.2.7 and you can solve for C , the viscometer constant.

$$\eta = C \rho t \quad (2.2.7)$$

Orifice Viscometers

Commonly found in the oil industry, orifice viscometers consist of a reservoir, an orifice, and a receiver. These viscometers report viscosity in units of efflux time as the measurement consists of measuring the time it takes for a given liquid to travel from the orifice to the receiver. These instruments are not accurate as the set-up does not ensure that the pressure on the liquid remains constant and there is energy lost to friction at the orifice. The most common types of these viscometer include Redwood, Engler, Saybolt, and Ford cup viscometers. A Saybolt viscometer is represented in Figure 2.2.3.

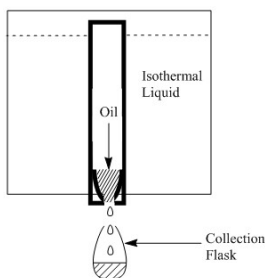


Figure 2.2.3 The time it takes for a 60 mL collection flask to fill is used to determine the viscosity in Saybolt units.

High Temperature, High Shear Rate Viscometers

These viscometers, also known as cylinder-piston type viscometers are employed when viscosities above 1000 poise, need to be determined, especially of non-Newtonian fluids. In a typical set-up, fluid in a cylindrical reservoir is displaced by a piston. As the pressure varies, this type of viscometry is well-suited for determining the viscosities over varying shear rates, ideal for characterizing fluids whose primary environment is a high temperature, high shear rate environment, e.g., motor oil. A typical cylinder-piston type viscometer is shown in Figure 2.2.4.

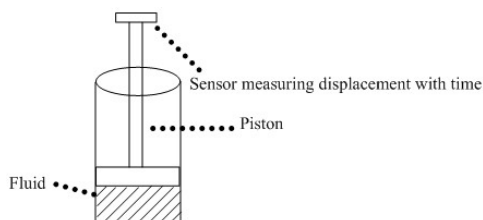


Figure 2.2.4 A typical cylinder-piston type viscometer.

Rotational Viscometers

Well-suited for non-Newtonian fluids, rotational viscometers measure the rate at which a solid rotates in a viscous medium. Since the rate of rotation is controlled, the amount of force necessary to spin the solid can be used to calculate the viscosity. They are advantageous in that a wide range of shear stresses and temperatures can be sampled across. Common rotational viscometers include: the coaxial-cylinder viscometer, cone and plate viscometer, and conical-cylinder viscometer. A cone and plate viscometer is shown in Figure 2.2.5.

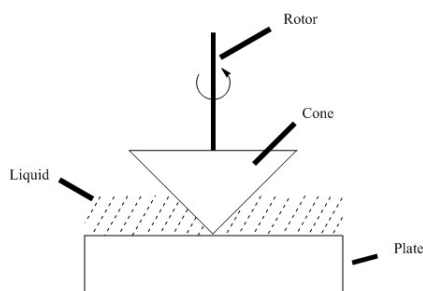


Figure 2.2.5 A cone is spun by a rotor in a liquid paste along a plate. The response of the rotation of the cone is measured, thereby determining viscosity.

Falling Ball Viscometer

This type of viscometer relies on the terminal velocity achieved by a ball falling through the viscous liquid whose viscosity is being measured. A sphere is the simplest object to be used because its velocity can be determined by rearranging Stokes' law Equation 2.2.8 to Equation 2.2.9, where r is the sphere's radius, η the dynamic viscosity, v the terminal velocity of the sphere, σ the density of the sphere, ρ the density of the liquid, and g the gravitational constant

$$6\pi r \eta v = \frac{4}{3} \pi r^3 (\sigma - \rho) g \quad (2.2.8)$$

$$\eta = \frac{\frac{4}{3} \pi r^2 (\sigma - \rho) g}{6\pi v} \quad (2.2.9)$$

A typical falling ball viscometric apparatus is shown in Figure 2.2.6.

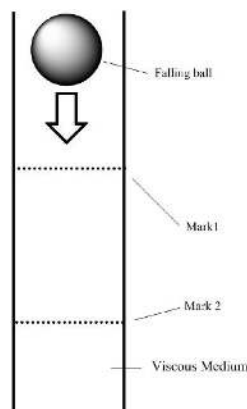


Figure 2.2.6 The time taken for the falling ball to pass from mark 1 to mark 2 is used to obtain viscosity measurements.

Vibrational Viscometers

Often used in industry, these viscometers are attached to fluid production processes where a constant viscosity quality of the product is desired. Viscosity is measured by the damping of an electrochemical resonator immersed in the liquid to be tested. The resonator is either a cantilever, oscillating beam, or a tuning fork. The power needed to keep the oscillator oscillating at a given frequency, the decay time after stopping the oscillation, or by observing the difference when waveforms are varied are respective ways in which this type of viscometer works. A typical vibrational viscometer is shown in Figure 2.2.7.

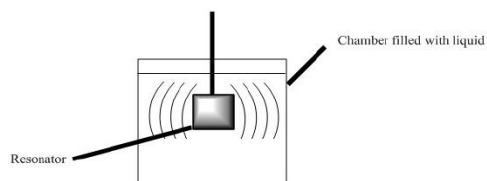


Figure 2.2.7 A resonator produces vibrations in the liquid whose viscosity is to be tested. An external sensor detects the vibrations with time, characterizing the material's viscosity in realtime.

This page titled 2.2: Viscosity Theory is shared under a CC BY 4.0 license and was authored, remixed, and/or curated by Pavan M. V. Raja & Andrew R. Barron (OpenStax CNX) via source content that was edited to the style and standards of the LibreTexts platform.

- 2.6: Viscosity by Pavan M. V. Raja & Andrew R. Barron is licensed CC BY 4.0. Original source: <http://cnx.org/contents/ba27839d-5042-4a40-afcf-c0e6e39fb45425.2>.

CHAPTER OVERVIEW

3: Surface Tension

3.1: What is surface tension?

3.2: Cohesive and Adhesive Forces

3.3: Contact Angles

3.4: Capillary Action

3: Surface Tension is shared under a [not declared](#) license and was authored, remixed, and/or curated by LibreTexts.

3.1: What is surface tension?

Surface tension is the energy, or work, required to increase the surface area of a liquid due to [intermolecular forces](#). Since these intermolecular forces vary depending on the nature of the liquid (e.g. water vs. gasoline) or solutes in the liquid (e.g. surfactants like detergent), each solution exhibits differing surface tension properties. Whether you know it or not, you already have seen surface tension at work. Whenever you fill a glass of water too far, you may notice afterward that the level of the water in the glass is actually higher than the height of the glass. You may have also noticed that the water that you spilled has formed into pools that rise up off the counter. Both of these phenomena are due to surface tension.



Water droplets form on the surface of the water pool because of surface tension.

Molecular Perspective

In a sample of water, there are two types of molecules. Those that are on the outside, exterior, and those that are on the inside, interior. The interior molecules are attracted to all the molecules around them, while the exterior molecules are attracted to only the other surface molecules and to those below the surface. This makes it so that the energy state of the molecules on the interior is much lower than that of the molecules on the exterior. Because of this, the molecules try to maintain a minimum surface area, thus allowing more molecules to have a lower energy state. This is what creates what is referred to as surface tension. An illustration of this can be seen in Figure 3.1.1.

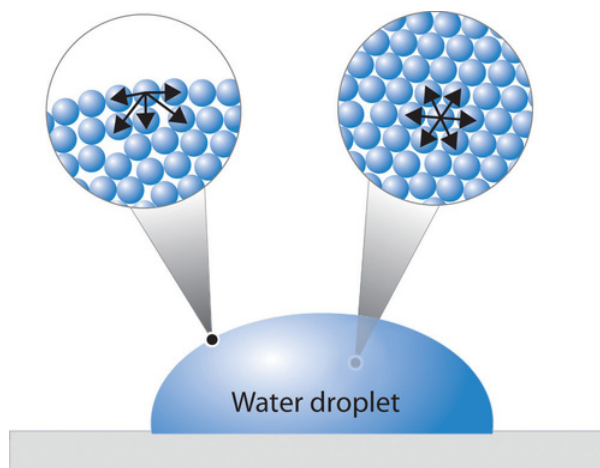


Figure 3.1.1: Molecules at the surface of water experience a net attraction to other molecules in the liquid, which holds the surface of the bulk sample together. In contrast, those in the interior experience uniform attractive forces. (CC BY-SA-NC; anonymous by request).

The water molecules attract one another due to the water's polar property. The hydrogen ends, which are positive in comparison to the negative ends of the oxygen cause water to "stick" together. This is why there is surface tension and takes a certain amount of energy to break these intermolecular bonds. Same goes for other liquids, even hydrophobic liquids such as oil. There are forces between the liquid such as Van der Waals forces that are responsible for the intermolecular forces found within the liquid. It will then take a certain amount of energy to break these forces, and the surface tension. Water is one liquid known to have a very high surface tension value and is difficult to overcome.

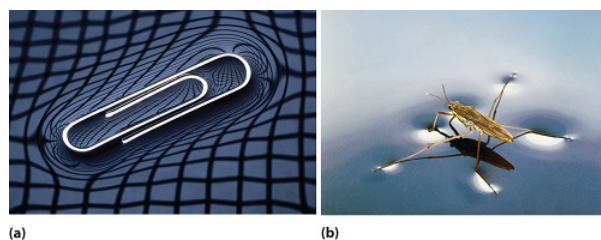


Figure 3.1.2: (a) A paper clip can “float” on water because of surface tension. (b) Surface tension also allows insects such as this water strider to “walk on water.” (CC BY-SA-NC; anonymous by request).

Surface tension of water can cause things to float which are denser than water, allowing organisms to literally walk on water (Figure 3.1.2). An example of such an organism is the water strider, which can run across the surface of water, due to the intermolecular forces of the molecules, and the force of the strider which is distributed to its legs. Surface tension also allows for the formation of droplets that we see in nature.

Cohesive and Adhesive Forces

There are several other important concepts that are related to surface tension. The first of these is the idea of [cohesive and adhesive Forces](#). Cohesive forces are those that hold the body of a liquid together with minimum surface area and adhesive forces are those that try to make a body of a liquid spread out. So if the cohesive forces are stronger than the adhesive forces, the body of water will maintain its shape, but if the opposite is true then the liquid will be spread out, *maximizing* its surface area. Any substance that you can add to a liquid that allows a liquid to increase its surface area is called a [wetting agent](#).

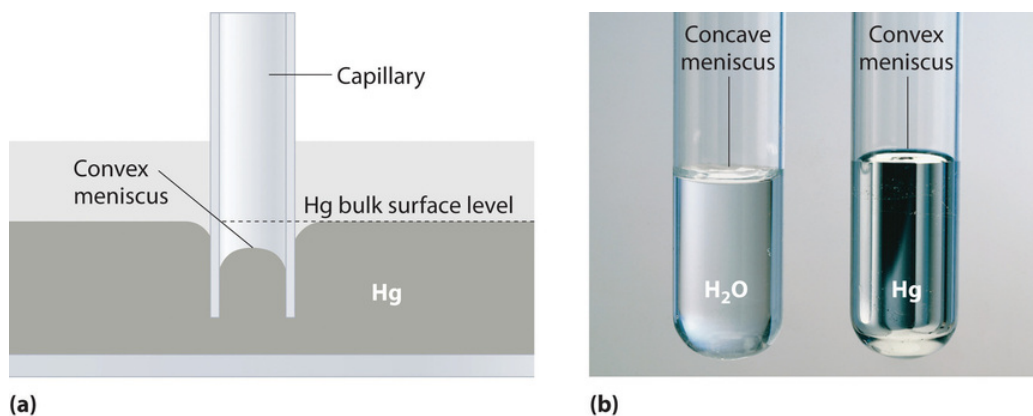


Figure 3.1.3: (a) This drawing illustrates the shape of the meniscus and the relative height of a mercury column when a glass capillary is put into liquid mercury. The meniscus is convex and the surface of the liquid inside the tube is lower than the level of the liquid outside the tube. (b) Because water adheres strongly to the polar surface of glass, it has a concave meniscus, whereas mercury, which does not adhere to the glass, has a convex meniscus. (CC BY-SA-NC; anonymous by request).

In the lab there are also several important points to remember about surface tension. The first you've probably noticed before. This is the idea of a meniscus (Figure 3.1.3a). This is the concave (curved in) or convex (curved out) look that water or other liquids have when they are in test tubes. This is caused by the attraction between the glass and the liquid. With water, this causes it to climb up the sides of a test tube. This attraction is amplified as the diameter of the tubes increases; this is called [capillary action](#). This can be seen if you take a tube with a very small diameter (a *capillary tube*) and lower it into a body of water. The liquid will climb up into the tube, even though there is no outside force. You may have also seen this when you put a straw into a drink and notice that the liquid level inside the straw is higher than it is in your drink. All of this however, requires that the adhesive forces (between the liquid and the capillary surface) be higher than the cohesive forces (between the liquid and itself), otherwise there will be no capillary action or the opposite can even happen. Mercury has higher cohesive forces than adhesive forces, so the level of the liquid will actually be lower in the capillary tubes than compared to the rest of the mercury (Figure 3.1.3b).

References

1. Petrucci, Ralph H., et al. General Chemistry: Principles and Modern Applications. Upper Saddle River, NJ: Prentice Hall, 2007.

Contributors and Attributions

- Benjamin Aldridge, Nivaz Brar

3.1: [What is surface tension?](#) is shared under a [not declared](#) license and was authored, remixed, and/or curated by LibreTexts.

3.2: Cohesive and Adhesive Forces

Cohesive and adhesive forces are associated with bulk (or macroscopic) properties, and hence the terms are not applicable to the discussion of [atomic and molecular properties](#). When a liquid comes into contact with a surface (such as the walls of a graduated cylinder or a tabletop), both cohesive and adhesive forces will act on it. These forces govern the shape which the liquid takes on. Due to the effects of adhesive forces, liquid on a surface can spread out to form a thin, relatively uniform film over the surface, a process known as [wetting](#). Alternatively, in the presence of strong cohesive forces, the liquid can divide into a number of small, roughly spherical beads that stand on the surface, maintaining minimal contact with the surface.

Adhesive and Cohesive Forces

The term "cohesive forces" is a generic term for the collective [intermolecular forces](#) (e.g., [hydrogen bonding](#) and [van der Waals forces](#)) responsible for the bulk property of liquids resisting separation. Specifically, these attractive forces exist between molecules of the *same* substance. For instance, rain falls in droplets, rather than a fine mist, because water has strong cohesion, which pulls its molecules tightly together, forming droplets. This force tends to unite molecules of a liquid, gathering them into relatively large clusters due to the molecules' dislike for its surroundings.

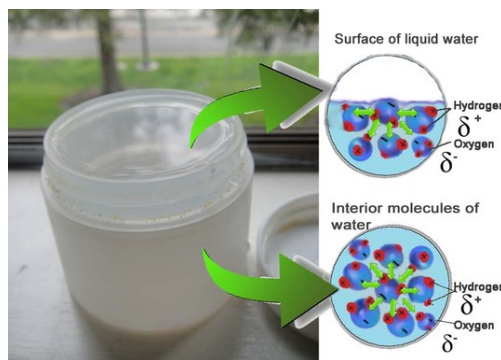
Similarly, the term "adhesive forces" refers to the attractive forces between unlike substance, such as mechanical forces (sticking together) and electrostatic forces (attraction due to opposing charges). In the case of a liquid wetting agent, adhesion causes the liquid to cling to the surface on which it rests. When water is poured on clean glass, it tends to spread, forming a thin, uniform film over the glass surface. This is because the adhesive forces between water and glass are strong enough to pull the water molecules out of their spherical formation and hold them against the surface of the glass, thus avoiding the repulsion between like molecules.

Macroscopic Effects of Cohesive and Adhesive Forces

When a liquid is placed on a smooth surface, the relative strengths of the cohesive and adhesive forces acting on that liquid determine the shape it will take (and whether or not it will wet the surface). If the adhesive forces between a liquid and a surface are stronger, they will pull the liquid down, causing it to wet the surface. However, if the cohesive forces among the liquid itself are stronger, they will resist such adhesion and cause the liquid to retain a spherical shape and bead the surface.

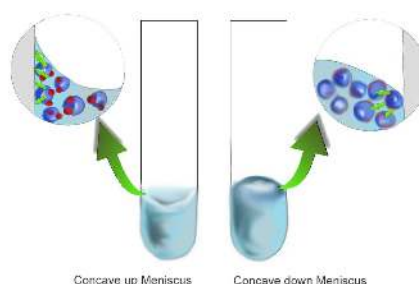
Case I: The Meniscus

The meniscus is the curvature of a liquid's surface within a container, such as a graduated cylinder. However, before we explain why some liquid have a concave up meniscus while others share a concave down meniscus, we have to understand the adhesive forces at work of [surface tension](#). Water, for example, is a polar molecule that consists of a partial positive charge on the hydrogens and a partial negative charge on the oxygen. Thus, within liquid water, each molecule's partial positive charge is attracted to its neighbor's partial negative charge. This is the origin of the cohesive forces within the water. Water molecules buried inside the liquid is then being pulled and pushed evenly in every direction, producing no net pull. Meanwhile, the molecules on the surface of the liquid, lacking pulling forces in the upward direction, thus encompass a net downward pull.



How does this cohesive force create both a concave up and concave down surface then? The answer is in its relationship to the adhesive forces between the water molecules and the container's surface. When the cohesive force of the liquid is stronger than the adhesive force of the liquid to the wall, the liquid concaves down in order to reduce contact with the surface of the wall. When the

adhesive force of the liquid to the wall is stronger than the cohesive force of the liquid, the liquid is more attracted to the wall than its neighbors, causing the upward concavity.



Case II: Tears of Wine

In agitated glasses of wine, droplets of wine seemingly "float" above the meniscus of the liquid and form "tears." This age-old phenomenon is the result of surface tension and cohesive and adhesive forces. Alcohol is more volatile than water. As a result, "evaporation of alcohol produces a surface tension gradient driving a thin film up along the side of a wine glass" (Adamson). This process is called the "solutal Marangoni effect."² Due to adhesive forces, some water clings to the walls of the glass. The "tears" form from the cohesive forces within the water holding it together. It is important to note that the surface tension gradient is "the driving force for the *motion* of the liquid" (Gugliotti), but the actual formation of the tears is a result of cohesive and adhesive forces.



The "tears of wine" are observed as a ring of clear liquid, near the top of a glass of wine, from which droplets continuously form and drop back into the wine. It is most readily observed in a wine which has a high alcohol content. (CC BY-SA-NC; [Greg Emel](#))

Problems

1. Name two examples where the cohesive force dominates over the adhesive force and vice versa.
2. When in a glass graduated cylinder, water presents an upwardly concave meniscus. However, when water is filled to the tip of the cylinder, the water level could maintain higher than the wall of the cylinder without pouring out resembling a concave down meniscus. Use the principles of cohesive and adhesive forces to explain this situation.
3. Explain why a water strider can glide on the water with the knowledge of cohesion in water.
4. Propose different types of forces that adhesive forces can build on.

Answers

1. When cohesive force is stronger than the adhesive force: concave up meniscus, water forms droplets on the surface. When adhesive force is stronger than the cohesive force: concave down meniscus, the surfaces are covered by the wetting agent, the last drops of liquid in the bottle always refuse to come out.
2. Since water forms a concave up meniscus, the adhesion of the molecules to the glass is stronger than the cohesion among the molecules. However, in the absence of the adhesive force (when water reaches the tip of the glass), the cohesive force remains present. Thus cohesive force alone proves that it can still hold itself in place without pouring out of the cylinder. This example emphasizes the importance of that cohesive force and adhesive forces do not simply cancel each other out, yet it is the difference between the two that determines the characteristic of the liquid.

3. This problem addresses once again the concept of surface tension. Because the cohesion of the water is built on the weak intermolecular forces of the water, when a water strider stepson to the surface, an extra energy will be necessary to overcome to break those bonds to increase surface area. Moreover, since the gravitational pull down on the water strider cannot overcome the activation energy to break these intermolecular forces, the water strider is able to glide freely on the water.
4. Complementary shape, chemical bonds form, weak intermolecular forces such as H-bonding or [Van der Waals](#) forces.

References

1. Petrucci, et al. General Chemistry: Principles & Modern Applications: AIE (Hardcover). Upper Saddle River: Pearson/Prentice Hall, 2007.
2. Gugliotti, Marcos. "Tears of Wine." *Journal of Chemical Education* 81.1 (2004): 67-68. Web. 9 Mar. 2010.
3. Adamson, A. W.; Gast, A. P. *Physical Chemistry of Surfaces*, 6th ed.; John Wiley and Sons: New York, 1997; p 371.

Contributors and Attributions

- Cameron Tracy, Ling Xie, Irene Lym, Henry Li

3.2: Cohesive and Adhesive Forces is shared under a [CC BY](#) license and was authored, remixed, and/or curated by LibreTexts.

3.3: Contact Angles

Contact angle is one of the common ways to measure the wettability of a surface or material. Wetting refers to the study of how a liquid deposited on a solid (or liquid) substrate spreads out or the ability of liquids to form boundary surfaces with solid states. The wetting, as mentioned before is determined by measuring the contact angle, which the liquid forms in contact with the solids or liquids. The wetting tendency is larger, the smaller the contact angle or the [surface tension](#) is. A wetting liquid is a liquid that forms a contact angle with the solid which is smaller than 90° . A non-wetting liquid creates a contact angle between 90° and 180° with the solid.

Introduction

The contact angle is an angle that a liquid creates with a solid surface or capillary walls of a porous material when both materials come in contact together. This angle is determined by both properties of the solid and the liquid and the interaction and repulsion forces between liquid and solid and by the three phase interface properties (gas, liquid and solid). Those interactions are described by [cohesion and adhesion](#) forces which are intermolecular forces. The balance between the cohesive forces of similar molecules such as between the liquid molecules (i.e. hydrogen bonds and Van der Waals forces) and the adhesive forces between dissimilar molecules such as between the liquid and solid molecules (i.e. mechanical and electrostatic forces) will determine the contact angle created in the solid and liquid interface. The traditional definition of a contact angle is the angle a liquid creates with the solid or liquid when it is deposited on it. A less traditional definition is the angle a liquid creates with the sides of a capillary when it rises in it to create a meniscus.

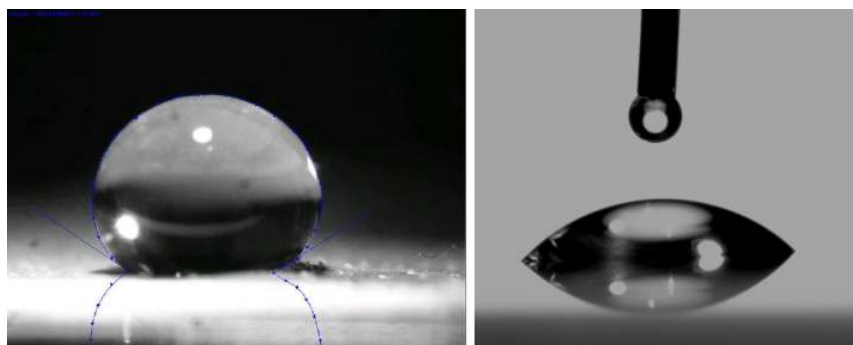


Figure 3.3.1: Images used with permission from Wikipedia

- The smaller the contact angle - cohesive forces are weaker than adhesive forces and molecules of the liquid tend to interact more with solid molecules than liquid molecules.
- The larger the contact angle - cohesive forces are stronger than adhesive forces and the molecules of the liquid tend to interact more with each other than with the solid molecules.

Thermodynamics

Thomas Young (13 June 1773 – 10 May 1829) proposed treating the contact angle of a liquid with a surface as the mechanical equilibrium of a drop resting on a plane solid surface under the restraints of three surface tensions:

- γ_{lv} (at the interface of the liquid and vapor phases),
- γ_{sl} (at the interface of the solid and liquid phases) and
- γ_{sv} (at the interface of the solid and vapor phase). This leads to *Young's equation*:

$$\gamma_{sv} - \gamma_{sl} = \gamma_{lv} \cos \theta \quad (3.3.1)$$

This equation is deceptively simple, but there are a few difficulties with it because of the definitions of the surface tension of the solid-vapor and solid-liquid phases.

Another approach avoids specifying the field of intermolecular forces between solid and liquid and instead offers a thermodynamic solution. This leads to Young and Dupre's equation which introduces the reversible work of adhesion of liquid and solid and its relation to the surface tension between liquid and vapor phases and the contact angle:

$$WA = \gamma_w(1 - \cos \theta) \quad (3.3.2)$$

Where WA is the reversible work of adhesion of the liquid to the solid when coated with an adsorbed film of the saturated vapor.

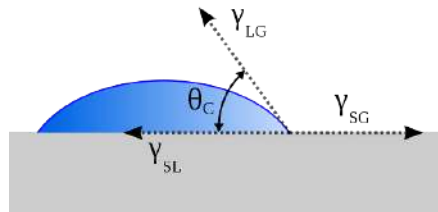


Figure 3.3.2: Contact action for the wetting of a surface

Less Traditional Definition

Capillarity is the ability of a substance to draw another substance into it. It occurs when the adhesive intermolecular forces between the liquid and a substance are stronger than the cohesive intermolecular forces inside the liquid. The effect causes a concave meniscus to form where the substance is touching a vertical surface (Figure 3.3.3). The same effect is what causes porous materials to soak up liquids. Capillary forces pull a wetting liquid has a low contact angle with the surface, it wets the surface. Assuming that there are no other factors involved (e.g. roughness), a low contact angle with water means that the surface is hydrophilic. A completely wetting liquid forms a zero contact angle into a capillary by creating a curved meniscus at the rising liquid front. This phenomenon can be described with the *Young-Laplace equation* and the Laplace pressure inside a capillary.

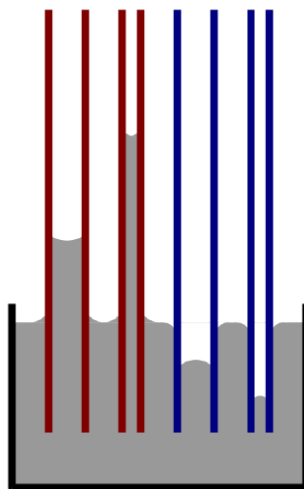


Figure 3.3.3: Illustration of [capillary rise](#) and contact angle in a capillary. Red=contact angle less than 90°; blue=contact angle greater than 90°. (Public Domain; [Eduard Konečný](#)).

Applications

The interest in contact angles is because it plays a significant role in a number of technological, environmental and biological phenomena. Water imbibition into porous media theory has been shown to have a multidisciplinary validity in food, soil physics, geology, printings and more. Imbibition of a liquid by a porous solid is a phenomenon highly dependent on wetting. Capillary imbibition is a mechanism that plays a significant role during rehydration of dry food particles that are considered as porous media. Imbibition is highly dependent on the wettability of the porous media, which is usually determined by measuring contact angles which liquids form with the solid.

Outside links

1. http://en.Wikipedia.org/wiki/Contact_angle
2. Contact angle measurements – The static and dynamic sessile drop method:
 1. www.youtube.com/watch?v=u265qIIUNrw

2. www.youtube.com/watch?v=5q9qOMesc88&feature=related
3. Equilibrium of water droplet on a hydrophobic surface:
 1. www.youtube.com/watch?v=HQAyyJsg17w&NR=1
4. Super hydrophobic materials:
 1. www.youtube.com/watch?v=EaE9k-xUtrQ&feature=related

References

1. Adamson, A.W. and Gast A.P. 1997. Physical Chemistry of Surfaces. 6th ed. John Wiley & Sons, New York. Pages 352-364.
2. Gould, R.F. 1964. Contact angle wettability and adhesion American Chemical Society, USA. Pages 1-4.
3. Seibold, A., Nardin, M., Schultz, J., Walliser, A. and Oppliger, M. 2000. Effect of dynamic contact angle on capillary rise phenomena. Colloids and Surfaces A: Physicochemical and Engineering Aspects, 161, 81-87.

Contributors and Attributions

- Keren Kles (UCD)

3.3: [Contact Angles](#) is shared under a [not declared](#) license and was authored, remixed, and/or curated by LibreTexts.

3.4: Capillary Action

Capillary action can be defined as the ascension of liquids through slim tube, cylinder or permeable substance due to adhesive and cohesive forces interacting between the liquid and the surface. When intermolecular bonding of a liquid itself is substantially inferior to a substances' surface it is interacting, capillarity occurs. Also, the diameter of the container as well as the gravitational forces will determine amount of liquid raised. While, water possesses this unique property, a liquid like mercury will not display the same attributes due to the fact that it has higher cohesive force than adhesive force.

Forces in Capillary Action

Three main variables that determine whether a liquid possesses capillary action are:

- **Cohesive force:** It is the intermolecular bonding of a substance where its mutual attractiveness forces them to maintain a certain shape of the liquid.
- **Surface tension:** This occurs as a result of like molecules, cohesive forces, banding together to form a somewhat impenetrable surface on the body of water. The surface tension is measured in milliNewton/meter (mN/m).
- **Adhesive force:** When forces of attraction between unlike molecules occur, it is called adhesive forces.

Capillary action only occurs when the adhesive forces are stronger than the cohesive forces, which invariably becomes [surface tension](#), in the liquid.

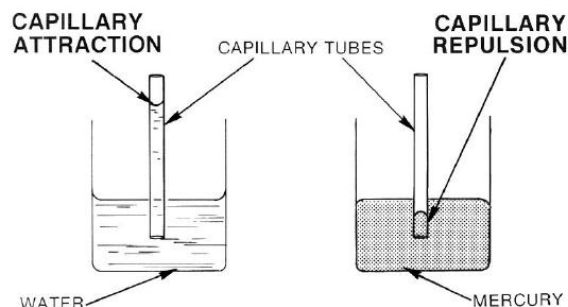


Figure 3.4.1: It is possible to see that in water, the strength of the adhesive forces are larger than the strength of the cohesive forces. This results in the concave formation of water in the capillary tube; this is known as capillary attraction. Alternatively for mercury, the cohesive forces are stronger than the adhesive forces which allows the the meniscus to bend away from the walls of the capillary tube. This is known as capillary Repulsion. [commons.wikimedia.org/wiki/File:\(PSF\)\(bjl\).svg](https://commons.wikimedia.org/wiki/File:(PSF)(bjl).svg)

A good way to remember the difference between adhesive and cohesive forces is that with *adhesive* forces you *add* another set of molecules, the molecules of the surface, for the liquid to bond with. With *cohesive* forces, the molecules of the liquid will only cooperate with their own kind. For water and clean glass, the adhesive forces (water/glass interface) are larger than the cohesive forces in water (hydrogen bonding) and the water wets the glass. For water on dirty glass, the adhesive forces (water/hydrocarbons (covering the glass surface)) can be smaller (depending on how dirty the glass surface is) than the cohesive forces in water resulting in droplets/beads of water on the glass surface. Larger adhesive than cohesive forces results in a rise up the capillary tube.

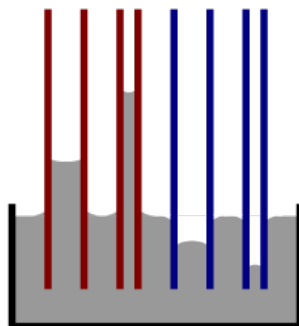


Figure 3.4.2: Scalable illustration of capillary action for large and small bore capillaries, and for positive and negative contact angles. (Public Domain; Eduard Konečný via Wikipedia).

Applications

Practical use of capillary action is evident in all forms of our daily lives. It makes performing our tasks efficiently and effectively. Some applications of this unique property include:

- The fundamental properties are used to absorb water by using paper towels. The cohesive and adhesive properties draw the fluid into the paper towel. The liquid flows into the paper towel at a certain rate.
- A technique called thin layer chromatography uses capillary action in which a layer of liquid is used to separate mixtures from substances.
- Capillary action helps us naturally by pumping out tear fluid in the eye. This process cleanses the eye and clears all of the dust and particles that are around the ducts of the eye.
- To generate energy: A possible use for capillary action is as a source of renewable energy. By allowing water to climb through capillaries, evaporate once it reaches the top, the condensate and drop back down to the bottom spinning a turbine on its way to create the energy, capillary action can make electricity! Although this idea is still in the works, it goes to show the potential that capillary action holds and how important it is.



Figure 3.4.3: Capillary action is evident in nature all around us. The properties allow the water to be transpired by the xylem in the plant. The water starts in the roots and proceeds upward to the highest branches of the plant.
commons.wikimedia.org/wiki/File:GemeineFichte.jpg

When measuring the level of liquid of a test tube or buret, it is imperative to measure at the meniscus line for an accurate reading. It is possible to measure the height (represented by h) of a test tube, buret, or other liquid column using the formula:

$$h = \frac{2\gamma \cos \theta}{\rho g r} \quad (3.4.1)$$

In this formula,

- γ represents the surface tension in a liquid-air environment,
- θ is the [angle of contact](#) or the degree of contact,
- ρ is the density of the liquid in the representative column,
- g is the acceleration due to the force of gravity and
- r is the radius of the tube in which the liquid is presented in.

By rearranging the equation and measuring the height of the capillary rise, h , surface tension can be determined.

$$\gamma = \frac{h \rho r}{2 \cos \theta} \quad (3.4.2)$$

Questions

1. Name one way to increase capillary action, and one way to decrease it.
2. If cohesion is greater than adhesion, will the meniscus be convex or concave?
3. What would be the height of water in a glass tube with a radius of .6mm? Assume the glass is clean (the contact angle between water and glass is 0°).

Solutions

1. Increase capillary action: Increase temperature, decrease capillary tube diameter, perform any number of actions to decrease surface tension, etc...! Decrease capillary action: The opposite of the steps you would take to increase, also, increasing the density of the liquid you're working with.
2. The meniscus will result in a convex formation.
3. The height of the water in the glass tube would be ~25mm high.

References

1. Petrucci, Ralph, and William Harwood. F. Geoffrey Herring. Jeffry Madura. General Chemistry: Principles and Modern Applications. 9th ed. Upper Saddle River, NJ: Pearson, 2007.
2. Cyclical Gravity Greenhouse Liquid Capillary Action Energy Generator, R Smith, 9.11.2005 14:38,
3. Capillary Action, Wikipedia
4. Capillary Action, from USGS Water Science for Schools." *USGS Georgia Water Science Center - Home Page*. Web. 04 June 2011.

Contributors and Attributions

- Achille Peiris, Becky Stein

3.4: Capillary Action is shared under a [not declared](#) license and was authored, remixed, and/or curated by LibreTexts.

CHAPTER OVERVIEW

4: MALDI-TOF

[4.1: Overview of mass spectrometry](#)

[4.2: Determining the formula of a fragment based on the mass of the Peak](#)

[4.3: The M+1 Peak](#)

4: MALDI-TOF is shared under a [not declared](#) license and was authored, remixed, and/or curated by LibreTexts.

4.1: Overview of mass spectrometry

Below is an introduction to mass spectrometry with a GC on the front end and a quadrupole mass spectrometer. In CHMY362 we will focus on the mass spectrometry of large molecules using Matrix Assisted Laser Desorption Ionization (MALDI) to get the molecules/fragments into the gas phase with a charge and a Time-of-Flight (ToF) mass spectrometer.

Gas Chromatography/Mass Spectrometry (GS/MS)

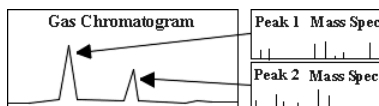
A mass spectrometer creates charged particles (ions) from molecules. It then analyzes those ions to provide information about the molecular weight of the compound and its chemical structure. There are many types of mass spectrometers and sample introduction techniques which allow a wide range of analyses. This discussion will focus on mass spectrometry as it's used in the powerful and widely used method of coupling Gas Chromatography (GC) with Mass Spectrometry (MS).



Pictured above is a GC/MS instrument used in organic teaching labs.

Gas Chromatograph (GC)

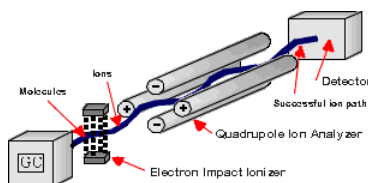
A mixture of compounds to be analyzed is initially injected into the GC where the mixture is vaporized in a heated chamber. The gas mixture travels through a GC column, where the compounds become separated as they interact with the column. The chromatogram on the right shows peaks which result from this separation. Those separated compounds then immediately enter the mass spectrometer.



Mass Spectrometer (MS)

Below is a general schematic of a mass spectrometer. The blue line illustrates ions of a particular mass/charge ratio which reach the detector at a certain voltage combination. All mass spectrometers consist of three distinct regions:

1. Ionizer
2. Ion Analyzer
3. Detector



Ionizer

In the GC-MS discussed in this introduction, the charged particles (ions) required for mass analysis are formed by Electron Impact (EI) Ionization. The gas molecules exiting the GC are bombarded by a high-energy electron beam (70 eV). An electron which strikes a molecule may impart enough energy to remove another electron from that molecule.

Ion Analyzer

Molecular ions and fragment ions are accelerated by manipulation of the charged particles through the mass spectrometer. Uncharged molecules and fragments are pumped away. The quadrupole mass analyzer in this example uses positive (+) and negative (-) voltages to control the path of the ions. Ions travel down the path based on their mass to charge ratio (m/z). EI ionization produces singly charged particles, so the charge (z) is one. Therefore an ion's path will depend on its mass. If the (+) and (-) rods shown in the mass spectrometer schematic were 'fixed' at a particular rf/dc voltage ratio, then one particular m/z would travel the successful path shown by the solid line to the detector. However, voltages are not fixed, but are scanned so that ever increasing masses can find a successful path through the rods to the detector.

Detector

There are many types of detectors, but most work by producing an electronic signal when struck by an ion. Timing mechanisms which integrate those signals with the scanning voltages allow the instrument to report which m/z strikes the detector. The mass analyzer sorts the ions according to m/z and the detector records the abundance of each m/z . Regular calibration of the m/z scale is necessary to maintain accuracy in the instrument. Calibration is performed by introducing a well known compound into the instrument and "tweaking" the circuits so that the compound's molecular ion and fragment ions are reported accurately.

Other Methods

An array of ionization methods and mass analyzers are available to meet the needs of many types of chemical analysis. A few are listed here with a highlight of their usefulness.

Sample introduction/ionization methods

Ionization method	Typical Analytes	Sample Introduction	Mass Range	Method Highlights
Electron Impact (EI)	Relatively small volatile	GC or liquid/solid probe	to 1,000 Daltons	Hard method versatile provides structure info
Chemical Ionization (CI)	Relatively small volatile	GC or liquid/solid probe	to 1,000 Daltons	Soft method molecular ion peak $[M+H]^+$
Electrospray (ESI)	Peptides Proteins nonvolatile	Liquid Chromatography or syringe	to 200,000 Daltons	Soft method ions often multiply charged
Fast Atom Bombardment (FAB)	Carbohydrates Organometallics Peptides nonvolatile	Sample mixed in viscous matrix	to 6,000 Daltons	Soft method but harder than ESI or MALDI
Matrix Assisted Laser Desorption (MALDI)	Peptides Proteins Nucleotides	Sample mixed in solid matrix	to 500,000 Daltons	Soft method very high mass

Mass Analyzers

Analyzer	System Highlights
Quadrupole	Unit mass resolution, fast scan, low cost
Sector (Magnetic and/or Electrostatic)	High resolution, exact mass
Time-of-Flight (TOF)	Theoretically, no limitation for m/z maximum, high throughput
Ion Cyclotron Resonance (ICR)	Very high resolution, exact mass, perform ion chemistry

Linked Systems

GC/MS:	Gas chromatography coupled to mass spectrometry
LC/MS:	Liquid chromatography coupled to electrospray ionization mass spectrometry

Outside Links

- A library of spectra can be found in the NIST WebBook, a data collection of the National Institute of Standards and Technology.
- Useful tools such as an exact mass calculator and a spectrum generator can be found in the MS Tools section of Scientific Instrument Services webpage.
- The JEOL Mass Spectrometry website contains tutorials, reference data and links to other sites.
- More general information and tutorials can be found in Scimedia, an educational resource.

Contributors and Attributions

Dr. Linda Brexi, Associate Director [Arizona Proteomics Consortium University of Arizona](#)

4.1: Overview of mass spectrometry is shared under a [not declared](#) license and was authored, remixed, and/or curated by LibreTexts.

4.2: Determining the formula of a fragment based on the mass of the Peak

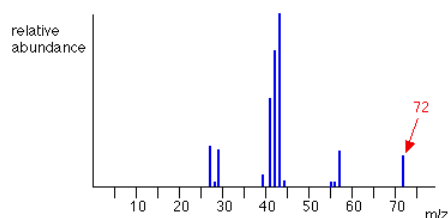
This page explains how to find the relative formula mass of a fragment from the peak at m/z in the mass spectrum. It also shows how high resolution mass spectra can be used to find the molecular formula for a compound.

Since we are using Matrix Assisted Laser Desorption (MALDI) to get large molecules into a gaseous phase, we will see either no fragmentation (mostly M^+ with some $M2^+$) or fragmentation of repeat units (polyethylene glycol fragments at the monomer units). Ionization can knock an electron off a molecule to form a positive ion. The molecular ion is often given the symbol M^+ or $M^{+\cdot}$ - the dot in this second version represents the fact that somewhere in the ion there will be a single unpaired electron. That's one half of what was originally a pair of electrons - the other half is the electron which was removed in the ionization process.

The example below is for a small organic molecule (molecular mass). The same principles for finding the mass of a fragment (and fragment formula) can be applied to the larger molecules (e.g. myoglobin) that we will use in lab.

In the mass spectrum, the heaviest ion (the one with the greatest m/z value) is likely to be the molecular ion. A few compounds have mass spectra which don't contain a molecular ion peak, because all the molecular ions break into fragments. For example, in the mass spectrum of pentane, the heaviest ion has an m/z value of 72.

simplified mass spectrum of pentane - $CH_3CH_2CH_2CH_2CH_3$



Because the largest m/z value is 72, that represents the largest ion going through the mass spectrometer - and you can reasonably assume that this is the molecular ion. The relative formula mass of the compound is therefore 72.

Finding the relative formula mass (relative molecular mass) from a mass spectrum is therefore trivial. Look for the peak with the highest value for m/z , and that value is the relative formula mass of the compound. There are, however, complications which arise because of the possibility of different isotopes (either of carbon or of chlorine or bromine) in the molecular ion. These cases are dealt with on separate pages.

Using a mass spectrum to find a molecular formula

So far we've been looking at m/z values in a mass spectrum as whole numbers, but it's possible to get far more accurate results using a high resolution mass spectrometer. You can use that more accurate information about the mass of the molecular ion to work out the molecular formula of the compound.

For normal calculation purposes, you tend to use rounded-off relative isotopic masses. For example, you are familiar with the atomic mass numbers (Z). To 4 decimal places, however, these are the relative isotopic masses.

Isotope	Z	Mass
1H	1	1.0078
^{12}C	12	12.0000
^{14}N	14	14.0031
^{16}O	16	15.9949

The carbon value is 12.0000, of course, because all the other masses are measured on the carbon-12 scale which is based on the carbon-12 isotope having a mass of exactly 12.

Using these accurate values to find a molecular formula

Two simple organic compounds have a relative formula mass of 44 - propane, C_3H_8 , and ethanal, CH_3CHO . Using a high resolution mass spectrometer, you could easily decide which of these you had. On a high resolution mass spectrometer, the molecular ion peaks for the two compounds give the following m/z values:

C_3H_8	44.0624
CH_3CHO	44.0261

You can easily check that by adding up numbers from the table of accurate relative isotopic masses above.

✓ Example 4.2.1:

A gas was known to contain only elements from the following list:

^1H	1.0078
^{12}C	12.0000
^{14}N	14.0031
^{16}O	15.9949

The gas had a molecular ion peak at $m/z = 28.0312$ in a high resolution mass spectrometer. What was the gas?

Solution

After a bit of playing around, you might reasonably come up with 3 gases which had relative formula masses of approximately 28 and which contained the elements from the list. They are N_2 , CO and C_2H_4 . Working out their accurate relative formula masses gives:

N_2	28.0062
CO	27.9949
C_2H_4	28.0312

The gas is obviously C_2H_4 .

Contributors and Attributions

- Jim Clark (Chemguide.co.uk)

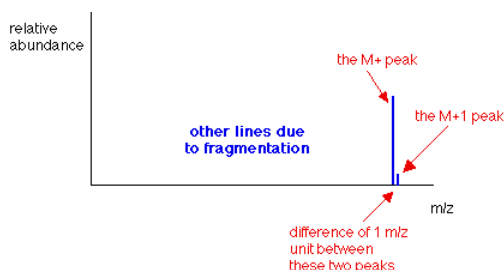
This page titled 4.2: Determining the formula of a fragment based on the mass of the Peak is shared under a CC BY-NC 4.0 license and was authored, remixed, and/or curated by Jim Clark.

4.3: The M+1 Peak

Any peak in a mass spectrum has a mass of M and a charge of z . Assuming a charge of $+1$ for a fragment with mass M , a peak at $M+1$ may be present in addition to the peak at M .

What causes the M+1 peak?

If you had a complete (rather than a simplified) mass spectrum, you will find a small line 1 m/z unit to the right of the main molecular ion peak. This small peak is called the $M+1$ peak.

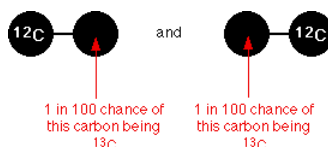


The carbon-13 isotope

The $M+1$ peak is caused by the presence of the ^{13}C isotope in the molecule. ^{13}C is a stable isotope of carbon - don't confuse it with the ^{14}C isotope which is radioactive. Carbon-13 makes up 1.11% of all carbon atoms. If you had a simple compound like methane, CH_4 , approximately 1 in every 100 of these molecules will contain carbon-13 rather than the more common carbon-12. That means that 1 in every 100 of the molecules will have a mass of 17 ($13 + 4$) rather than 16 ($12 + 4$). The mass spectrum will therefore have a line corresponding to the molecular ion $[^{13}\text{CH}_4]^+$ as well as $[^{12}\text{CH}_4]^+$. The line at $m/z = 17$ will be much smaller than the line at $m/z = 16$ because the carbon-13 isotope is much less common. Statistically you will have a ratio of approximately 1 of the heavier ions to every 99 of the lighter ones. That's why the $M+1$ peak is much smaller than the M peak.

Using the M+1 peak

Imagine a compound containing 2 carbon atoms. Either of them has an approximately 1 in 100 chance of being ^{13}C .



There's therefore a 2 in 100 chance of the molecule as a whole containing one ^{13}C atom rather than a ^{12}C atom - which leaves a 98 in 100 chance of both atoms being ^{12}C . That means that the ratio of the height of the $M+1$ peak to the M peak will be approximately 2 : 98. That's pretty close to having an $M+1$ peak approximately 2% of the height of the M peak.

Using the relative peak heights to predict the number of carbon atoms

If you measure the peak height of the $M+1$ peak as a percentage of the peak height of the M peak, that gives you the number of carbon atoms in the compound. We've just seen that a compound with 2 carbons will have an $M+1$ peak approximately 2% of the height of the M peak. Similarly, you could show that a compound with 3 carbons will have the $M+1$ peak at about 3% of the height of the M peak. The approximations we are making won't hold with more than 2 or 3 carbons. The proportion of carbon atoms which are ^{13}C isn't 1% - it's 1.11%. And the approximation that a ratio of 2 : 98 is about 2% doesn't hold as the small number increases.

Consider a molecule with 5 carbons in it. You could work out that 5.55 (5×1.11) molecules will contain 1 ^{13}C to every 94.45 ($100 - 5.55$) which contain only ^{12}C atoms. If you convert that to how tall the $M+1$ peak is as a percentage of the M peak, you get an answer of 5.9% ($5.55/94.45 \times 100$). That's close enough to 6% that you might assume wrongly that there are 6 carbon atoms. Above 3 carbon atoms, then, you shouldn't really be making the approximation that the height of the $M+1$ peak as a percentage of the height of the M peak tells you the number of carbons - you will need to do some fiddly sums!

We will not determine the number of carbons in a fragment based on the height ratio of M^+ to $M+1$, but the smaller peaks should be reasonable.

Other Isotopes

^{13}C has a relative abundance of $\sim 1\%$. Notice that other elements (^{34}S , ^{37}Cl and ^{81}Br) with high relative abundances could be very helpful when determining a fragment formula.

Relative Isotope Abundance of Common Elements

Element	Isotope	Relative Abundance	Isotope	Relative Abundance	Isotope	Relative Abundance
Carbon	^{12}C	100	^{13}C	1.11		
Hydrogen	^1H	100	^2H	0.016		
Nitrogen	^{14}N	100	^{15}N	0.38		
Oxygen	^{16}O	100	^{17}O	0.04	^{18}O	0.20
Sulfur	^{32}S	100	^{33}S	0.78	^{34}S	4.40
Chlorine	^{35}Cl	100			^{37}Cl	32.5
Bromine	^{79}Br	100			^{81}Br	98.0

M

Contributors and Attributions

Jim Clark (Chemguide.co.uk)

This page titled [4.3: The \$M+1\$ Peak](#) is shared under a [CC BY-NC 4.0](#) license and was authored, remixed, and/or curated by [Jim Clark](#).

CHAPTER OVERVIEW

5: Fluorometry

5.1: Fluorescence and Phosphorescence

5: Fluorometry is shared under a [not declared](#) license and was authored, remixed, and/or curated by LibreTexts.

5.1: Fluorescence and Phosphorescence

Fluorescence and phosphorescence are types of molecular luminescence. A molecule of analyte absorbs a photon and results in an excited species. That excited species can then emit energy (photon, heat). The emission spectrum can provide qualitative and quantitative analysis. The term fluorescence and phosphorescence are usually referred to as photoluminescence because both are alike in excitation brought by absorption of a photon. Fluorescence differs from phosphorescence in that the electronic energy transition that is responsible for fluorescence does not change in electron spin, which results in short-live electrons ($<10^{-5}$ s) in the excited state of fluorescence. In phosphorescence, there is a change in electron spin, which results in a longer lifetime of the excited state (second to minutes). Fluorescence and phosphorescence occur at longer wavelength than the excitation radiation.

Introduction

Fluorescence can occur in gaseous, liquid, and solid chemical systems. The simple kind of fluorescence is of dilute atomic vapors. A fluorescence example would be if a 3s electron of a vaporized sodium atom is excited to the 3p state by absorption of a radiation at wavelength 589.6 and 589.0 nm. After 10^{-8} s, the electron returns to ground state and on its return it emits radiation of the two wavelengths in all directions. This type of fluorescence in which the absorbed radiation is remitted without a change in frequency is known as resonance fluorescence. Resonance fluorescence can also occur in molecular species. Molecular fluorescence band centers at wavelengths longer than resonance lines. The shift toward longer wavelength is referred to as the Stokes Shift.

Singlet and Triplet Excited State

Understanding the difference between fluorescence and phosphorescence requires the knowledge of electron spin and the differences between singlet and triplet states. [The Pauli Exclusion principle](#) states that two electrons in an atom cannot have the same four quantum numbers (n, l, m_l, m_s) and only two electrons can occupy each orbital where they must have opposite spin states. These opposite spin states are called spin pairing. Because of this spin pairing, most molecules do not exhibit a magnetic field and are diamagnetic. In diamagnetic molecules, electrons are not attracted or repelled by the static electric field. Free radicals are paramagnetic because they contain unpaired electrons have magnetic moments that are attracted to the magnetic field.

Singlet state is defined when all the electron spins are paired in the molecular electronic state and the electronic energy levels do not split when the molecule is exposed into a magnetic field. A doublet state occurs when there is an unpaired electron that gives two possible orientations when exposed in a magnetic field and imparts different energy to the system. A singlet or a triplet can form when one electron is excited to a higher energy level. In an excited singlet state, the electron is promoted in the same spin orientation as it was in the ground state (paired). In a triplet excited state, the electron that is promoted has the same spin orientation (parallel) to the other unpaired electron. The difference between the spins of ground singlet, excited singlet, and excited triplet is shown in Figure 5.1.1. Singlet, doublet and triplet is derived using the equation for multiplicity, $2S+1$, where S is the total spin angular momentum (sum of all the electron spins). Individual spins are denoted as spin up ($s = +1/2$) or spin down ($s = -1/2$). If we were to calculate the S for the excited singlet state, the equation would be $2(+1/2 + -1/2)+1 = 2(0)+1 = 1$, therefore making the center orbital in the figure a singlet state. If the spin multiplicity for the excited triplet state was calculated, we obtain $2(+1/2 + +1/2)+1 = 2(1)+1 = 3$, which gives a triplet state as expected.

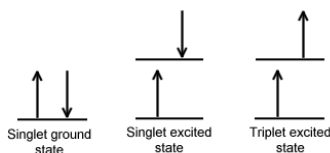


Figure 5.1.1: Spin in the ground and excited states.

The difference between a molecule in the ground and excited state is that the electrons is diamagnetic in the ground state and paramagnetic in the triplet state. This difference in spin state makes the transition from singlet to triplet (or triplet to singlet) more improbable than the singlet-to-singlet transitions. This singlet to triplet (or reverse) transition involves a change in *electronic* state. For this reason, the lifetime of the triplet state is longer the singlet state by approximately 10^4 seconds fold difference. The radiation that induced the transition from ground to excited triplet state has a low probability of occurring, thus their absorption bands are less intense than singlet-singlet state absorption. The excited triplet state can be populated from the excited singlet state of certain

molecules which results in phosphorescence. These spin multiplicities in ground and excited states can be used to explain transition in photoluminescence molecules by the Jablonski diagram.

Jablonski Diagrams

The Jablonski diagram that drawn below is a partial energy diagram that represents the energy of photoluminescent molecule in its different energy states. The lowest and darkest horizontal line represents the ground-state electronic energy of the molecule which is the singlet state labeled as S_0 . At room temperature, majority of the molecules in a solution are in this state.

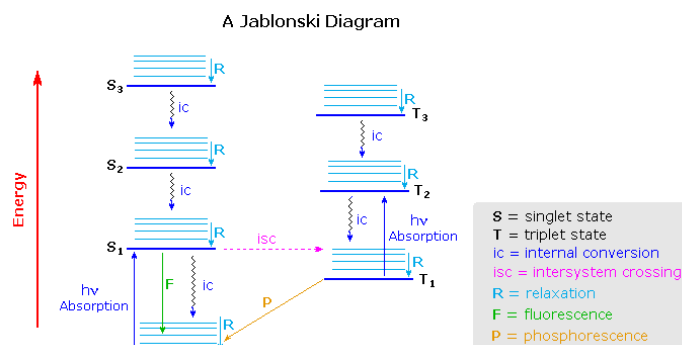


Figure 5.1.2: Partial Jablonski Diagram for Absorption, Fluorescence, and Phosphorescence. from Bill Reusch.

The upper lines represent the energy state of the three excited electronic states: S_1 and S_2 represent the electronic singlet state (left) and T_1 represents the first electronic triplet state (right). The upper darkest line represents the ground vibrational state of the three excited electronic state. The energy of the triplet state is lower than the energy of the corresponding singlet state.

There are numerous vibrational levels that can be associated with each electronic state as denoted by the thinner lines. Absorption transitions (blue lines in Figure 5.1.2) can occur from the ground singlet electronic state (S_0) to various vibrational levels in the singlet excited vibrational states. It is unlikely that a transition from the ground singlet electronic state to the triplet electronic state because the electron spin is parallel to the spin in its ground state (Figure 5.1.1). This transition leads to a change in multiplicity and thus has a low probability of occurring which is a **forbidden transition**. Molecules also go through vibration relaxation to lose any excess vibrational energy that remains when excited to the electronic states (S_1 and S_2) as demonstrated in wavy lines in Figure 5.1.2. The knowledge of forbidden transition is used to explain and compare the peaks of absorption and emission.

Absorption and Emission Rates

The table below compares the absorption and emission rates of fluorescence and phosphorescence. The rate of photon absorption is very rapid. Fluorescence emission occurs at a slower rate. Since the triplet to singlet (or reverse) is a forbidden transition, meaning it is less likely to occur than the singlet-to-singlet transition, the rate of triplet to singlet is typically slower. Therefore, phosphorescence emission requires more time than fluorescence.

Table 5.1.1: Rates of Absorption and Emission comparison.

Process	Transition	Timescale (sec)
Light Absorption (Excitation)	$S_0 \rightarrow S_n$	ca. 10^{-15} (instantaneous)
Internal Conversion	$S_n \rightarrow S_1$	10^{-14} to 10^{-11}
Vibrational Relaxation	$S_n^* \rightarrow S_n$	10^{-12} to 10^{-10}
Intersystem Crossing	$S_1 \rightarrow T_1$	10^{-11} to 10^{-6}
Fluorescence	$S_1 \rightarrow S_0$	10^{-9} to 10^{-6}
Phosphorescence	$T_1 \rightarrow S_0$	10^{-3} to 100
Non-Radiative Decay	$S_1 \rightarrow S_0$	10^{-7} to 10^{-5}
	$T_1 \rightarrow S_0$	10^{-3} to 100

Deactivation Processes

A molecule that is excited can return to the ground state by several combinations of mechanical steps that will be described below and shown in Figure 5.1.2. The deactivation process of fluorescence and phosphorescence involve an emission of a photon radiation as shown by the straight arrow in Figure 5.1.2. The wiggly arrows in Figure 5.1.2 are deactivation processes without the use of radiation. The favored deactivation process is the route that is most rapid and spends less time in the excited state. If the rate constant for fluorescence is more favorable in the radiationless path, the fluorescence will be less intense or absent.

- **Vibrational Relaxation:** A molecule may be promoted to several vibrational levels during the electronic excitation process. Collision of molecules with the excited species and solvent leads to rapid energy transfer and a slight increase in temperature of the solvent. Vibrational relaxation is so rapid that the lifetime of a vibrational excited molecule ($<10^{-12}$) is less than the lifetime of the electronically excited state. For this reason, fluorescence from a solution always involves the transition of the lowest vibrational level of the excited state. Since the space of the emission lines are so close together, the transition of the vibrational relaxation can terminate in any vibrational level of the ground state.
- **Internal Conversion:** Internal conversion is an intermolecular process of molecule that passes to a lower electronic state without the emission of radiation. It is a crossover of two states with the same multiplicity meaning singlet-to-singlet or triplet-to-triplet states. The internal conversion is more efficient when two electronic energy levels are close enough that two vibrational energy levels can overlap as shown in between S_1 and S_2 . Internal conversion can also occur between S_0 and S_1 from a loss of energy by fluorescence from a higher excited state, but it is less probable. The mechanism of internal conversion from S_1 to S_0 is poorly understood. For some molecules, the vibrational levels of the ground state overlaps with the first excited electronic state, which leads to fast deactivation. These usually occur with aliphatic compounds (compound that do not contain ring structure), which would account for the compound is seldom fluorescing. Deactivation by energy transfer of these molecules occurs so rapidly that the molecule does not have time to fluoresce.
- **External Conversion:** Deactivation of the excited electronic state may also involve the interaction and energy transfer between the excited state and the solvent or solute in a process called **external conversion**. Low temperature and high viscosity leads to enhanced fluorescence because they reduce the number of collision between molecules, thus slowing down the deactivation process.
- **Intersystem Crossing:** Intersystem crossing is a process where there is a crossover between electronic states of different multiplicity as demonstrated in the singlet state to a triplet state (S_1 to T_1) on Figure 5.1.1. The probability of intersystem crossing is enhanced if the vibration levels of the two states overlap. Intersystem crossing is most commonly observed with molecules that contain heavy atom such as iodine or bromine. The spin and orbital interaction increase and the spin become more favorable. Paramagnetic species also enhances intersystem crossing, which consequently decreases fluorescence.
- **Phosphorescence:** Deactivation of the electronic excited state is also involved in phosphorescence. After the molecule transitions through intersystem crossing to the triplet state, further deactivation occurs through internal or external fluorescence or phosphorescence. A triplet-to-singlet transition is more probable than a singlet-to-singlet internal crossing. In phosphorescence, the excited state lifetime is inversely proportional to the probability that the molecule will transition back to the ground state. Since the lifetime of the molecule in the triplet state is large (10^{-4} to 10 second or more), transition is less probable which suggest that it will persist for some time even after irradiation has stopped. Since the external and internal conversion compete so effectively with phosphorescence, the molecule has to be observed at lower temperature in highly viscous media to protect the triplet state.

Variables that affect Fluorescence

After discussing all the possible deactivation processes, variable that affect the emissions to occur. Molecular structure and its chemical environment influence whether a substance will fluoresce and the intensities of these emissions. The **quantum yield** or **quantum efficiency** is used to measure the probability that a molecule will fluoresce or phosphoresce. For fluorescence and phosphorescence is the ratio of the number of molecules that luminescent to the total number of excited molecules. For highly fluoresce molecules, the quantum efficiency approaches to one. Molecules that do not fluoresce have quantum efficiencies that approach to zero.

Fluorescence quantum yield (ϕ) for a compound is determined by the relative rate constants (k) of various deactivation processes by which the lowest excited singlet state is deactivated to the ground state. The deactivation processes including fluorescence (k_f), intersystem crossing (k_i), internal conversion (k_{ic}), predissociation (k_{pd}), dissociation (k_d), and external conversion (k_{ec}) allows

one to qualitatively interpret the structural and environmental factors that influence the intensity of the fluorescence. They are related by the quantum yield equation given below:

$$\frac{k_f}{k_f + k_i + k_{ec} + k_{ic} + k_{pd} + k_d} \quad (5.1.1)$$

Using this equation as an example to explain fluorescence, a high fluorescence rate (k_f) value and low values of the all the other relative rate constant terms ($k_f + k_i + k_{ec} + k_{ic} + k_{pd} + k_d$) will give a large ϕ , which suggest that fluorescence is enhanced. The magnitudes of k_f , k_d , and k_{pd} depend on the chemical structure, while the rest of the constants k_i , k_{ec} , and k_{ic} are strongly influenced by the environment.

Molecules that are excited electronically will return to the lowest excited state by rapid vibrational relaxation and internal conversion, which produces no radiation emission. Fluorescence arises from a transition from the lowest vibrational level of the first excited electronic state to one of the vibrational levels in the electronic ground state. In most fluorescent compounds, radiation is produced by a $\pi^* \rightarrow \pi$ or $\pi^* \rightarrow n$ transition depending on which requires the least energy for the transition to occur.

Fluorescence is most commonly found in compounds in which the lowest energy transition is $\pi \rightarrow \pi^*$ (excited singlet state) than $n \rightarrow \pi^*$ which suggest that the quantum efficiency is greater for $\pi \rightarrow \pi^*$ transitions. The reason for this is that the molar absorptivity, which measures the probability that a transition will occur, of the $\pi \rightarrow \pi^*$ transition is 100 to 1000 fold greater than $n \rightarrow \pi^*$ process. The lifetime of $\pi \rightarrow \pi^*$ (10^{-7} to 10^{-9} s) is shorter than the lifetime of $n \rightarrow \pi^*$ (10^{-5} to 10^{-7}).

Quenching Methods

Dynamic Quenching is a nonradiative energy transfer between the excited and the quenching agent species (Q). The requirements for a successful dynamic quenching are that the two collision species the concentration must be high so that there is a higher possibility of collision between the two species. Temperature and quenching agent viscosity play a role on the rate of dynamic quenching. Dynamic quenching reduces fluorescence quantum yield and the fluorescence lifetime.

Dissolved oxygen in a solution increases the intensity of the fluorescence by photochemically inducing oxidation of the fluorescing species. Quenching results from the paramagnetic properties of molecular oxygen that promotes intersystem crossing and converts the excited molecules to triplet state. Paramagnetic species and dissolved oxygen tend to quench fluorescence and quench the triplet state.

Static quenching occurs when the quencher and ground state fluorophore forms a dark complex. Fluorescence is usually observed from unbound fluorophore. Static quenching can be differentiated from dynamic quenching in that the lifetime is not affected in static quenching. In long range (Förster) quenching, energy transfer occurs without collision between molecules, but dipole-dipole coupling occurs between excited fluorophore and quencher.

Emission and Excitation Spectra

One of the ways to visually distinguish the difference between each photoluminescence is to compare the relative intensities of emission/excitation at each wavelength. An example of the three types of photoluminescence (absorption, fluorescence and phosphorescence) is shown for phenanthrene in the spectrum below. In the spectrum, the luminescent intensity is measure in a wavelength is fixed while the excitation wavelength is varied. The spectrum in red represents the excitation spectrum, which is identical to the absorption spectrum because in order for fluorescence emission to occur, radiation needs to be absorbed to create an excited state. The spectrum in blue represent fluorescence and green spectrum represents the phosphorescence.

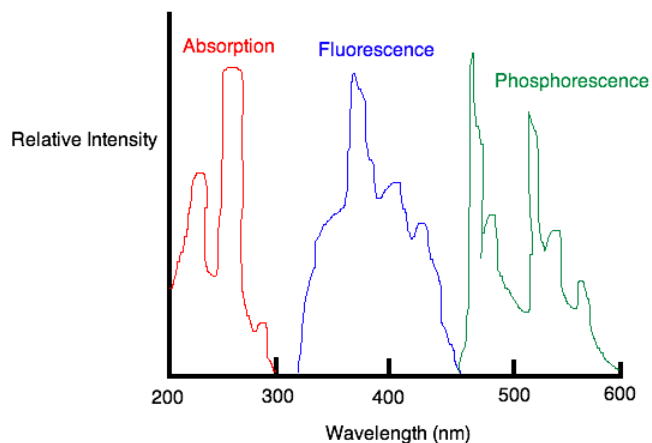


Figure 3: Wavelength Intensities of Absorption, Fluorescence, and Phosphorescence

Fluorescence and Phosphorescence occur at wavelengths that are longer than their absorption wavelengths. Phosphorescence bands are found at a longer wavelength than fluorescence band because the excited triplet state is lower in energy than the singlet state. The difference in wavelength could also be used to measure the energy difference between the singlet and triplet state of the molecule. The wavelength (λ) of a molecule is inversely related to the energy (E) by the equation below:

$$E = \frac{hc}{\lambda} \quad (5.1.2)$$

As the wavelength increases, the energy of the molecule decrease and vice versa.

References

1. D. A. Skoog, et al. "Principles of Instrumental Analysis" 6th Edition, Thomson Brooks/Cole. 2007
2. D. C. Harris and M.D. Bertolucci "Symmetry and Spectroscopy, An Introduction to Vibrational and Electronic Spectroscopy" Dover Publications, Inc., New York. 1989.

Contributors and Attributions

- Diana Wong (UCD)

5.1: Fluorescence and Phosphorescence is shared under a [CC BY 4.0](#) license and was authored, remixed, and/or curated by LibreTexts.

- [Fluorescence and Phosphorescence](#) by Diana Wong is licensed [CC BY 4.0](#).

CHAPTER OVERVIEW

6: UV-Vis

6.1: What Causes Molecules to Absorb UV and Visible Light

6.2: Solar Energy - UVA, UVB, UVC

6.3: Sunscreen

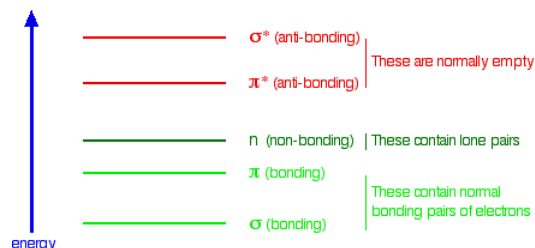
6: UV-Vis is shared under a [not declared](#) license and was authored, remixed, and/or curated by LibreTexts.

6.1: What Causes Molecules to Absorb UV and Visible Light

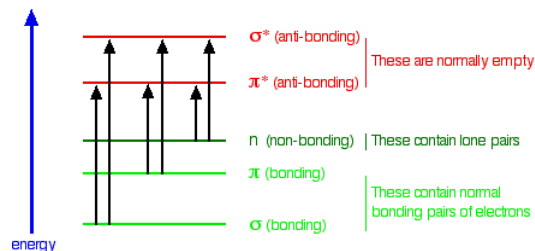
This page explains what happens when organic compounds absorb UV or visible light, and why the wavelength of light absorbed varies from compound to compound.

What happens when light is absorbed by molecules?

When we were talking about the various sorts of orbitals present in organic compounds on the introductory page (see above), you will have come across this diagram showing their relative energies:



Remember that the diagram isn't intended to be to scale - it just shows the relative placing of the different orbitals. When light passes through the compound, energy from the light is used to promote an electron from a bonding or non-bonding orbital into one of the empty anti-bonding orbitals. The possible electron jumps that light might cause are:



In each possible case, an electron is excited from a full orbital into an empty anti-bonding orbital. Each jump takes energy from the light, and a big jump obviously needs more energy than a small one. Each wavelength of light has a particular energy associated with it. If that particular amount of energy is just right for making one of these energy jumps, then that wavelength will be absorbed - its energy will have been used in promoting an electron.

We need to work out what the relationship is between the energy gap and the wavelength absorbed. Does, for example, a bigger energy gap mean that light of a lower wavelength will be absorbed - or what? It is easier to start with the relationship between the frequency of light absorbed and its energy:

$$E = h\nu$$

energy of the light frequency of the light
Pland's constant

You can see that if you want a high energy jump, you will have to absorb light of a higher frequency. The greater the frequency, the greater the energy. That's easy - but unfortunately UV-visible absorption spectra are always given using wavelengths of light rather than frequency. That means that you need to know the relationship between wavelength and frequency.

$$\lambda = \frac{c}{\nu}$$

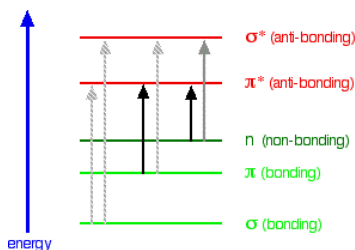
wavelength speed of light
frequency

You can see from this that the higher the frequency is, the lower the wavelength is. So, if you have a bigger energy jump, you will absorb light with a higher frequency - which is the same as saying that you will absorb light with a lower wavelength.

Important summary: The larger the energy jump, the lower the wavelength of the light absorbed.

Some jumps are more important than others for absorption spectrometry

An absorption spectrometer works in a range from about 200 nm (in the near ultra-violet) to about 800 nm (in the very near infra-red). Only a limited number of the possible electron jumps absorb light in that region. Look again at the possible jumps. This time, the important jumps are shown in black, and a less important one in grey. The grey dotted arrows show jumps which absorb light outside the region of the spectrum we are working in.



Remember that bigger jumps need more energy and so absorb light with a shorter wavelength. The jumps shown with grey dotted arrows absorb UV light of wavelength less than 200 nm. The important jumps are:

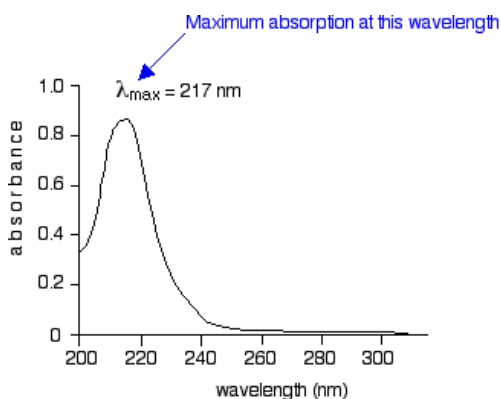
- from pi bonding orbitals to pi anti-bonding orbitals;
- from non-bonding orbitals to pi anti-bonding orbitals;
- from non-bonding orbitals to sigma anti-bonding orbitals.

That means that in order to absorb light in the region from 200 - 800 nm (which is where the spectra are measured), the molecule must contain either pi bonds or atoms with non-bonding orbitals. Remember that a non-bonding orbital is a lone pair on, say, oxygen, nitrogen or a halogen.

Groups in a molecule which absorb light are known as chromophores.

What does an absorption spectrum look like

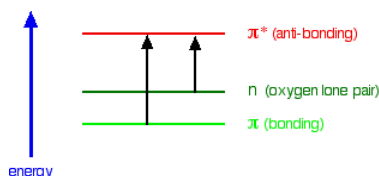
The diagram below shows a simple UV-visible absorption spectrum for buta-1,3-diene - a molecule we will talk more about later. Absorbance (on the vertical axis) is just a measure of the amount of light absorbed. The higher the value, the more of a particular wavelength is being absorbed.



You will see that absorption peaks at a value of 217 nm. This is in the ultra-violet and so there would be no visible sign of any light being absorbed - buta-1,3-diene is colorless. You read the symbol on the graph as "lambda-max". In buta-1,3-diene, $\text{CH}_2=\text{CH}-\text{CH}=\text{CH}_2$, there are no non-bonding electrons. That means that the only electron jumps taking place (within the range that the spectrometer can measure) are from pi bonding to pi anti-bonding orbitals.

A chromophore producing two peaks

A chromophore such as the carbon-oxygen double bond in ethanal, for example, obviously has pi electrons as a part of the double bond, but also has lone pairs on the oxygen atom. That means that both of the important absorptions from the last energy diagram are possible. You can get an electron excited from a pi bonding to a pi anti-bonding orbital, or you can get one excited from an oxygen lone pair (a non-bonding orbital) into a pi anti-bonding orbital.



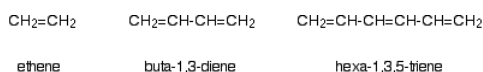
The non-bonding orbital has a higher energy than a pi bonding orbital. That means that the jump from an oxygen lone pair into a pi anti-bonding orbital needs less energy. That means it absorbs light of a lower frequency and therefore a higher wavelength. Ethanal can therefore absorb light of two different wavelengths:

- the pi bonding to pi anti-bonding absorption peaks at 180 nm;
- the non-bonding to pi anti-bonding absorption peaks at 290 nm.

Both of these absorptions are in the ultra-violet, but most spectrometers won't pick up the one at 180 nm because they work in the range from 200 - 800 nm.

The importance of conjugation and delocalisation

Consider these three molecules:



Ethene contains a simple isolated carbon-carbon double bond, but the other two have conjugated double bonds. In these cases, there is delocalization of the pi bonding orbitals over the whole molecule. Now look at the wavelengths of the light which each of these molecules absorbs.

molecule	wavelength of maximum absorption (nm)
ethene	171
buta-1,3-diene	217
hexa-1,3,5-triene	258

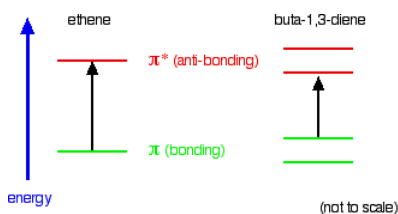
All of the molecules give similar UV-visible absorption spectra - the only difference being that the absorptions move to longer and longer wavelengths as the amount of delocalization in the molecule increases.

Why is this? You can actually work out what must be happening.

- The maximum absorption is moving to longer wavelengths as the amount of delocalization increases.
- Therefore maximum absorption is moving to shorter frequencies as the amount of delocalization increases.
- Therefore absorption needs less energy as the amount of delocalization increases.
- Therefore there must be less energy gap between the bonding and anti-bonding orbitals as the amount of delocalization increases.

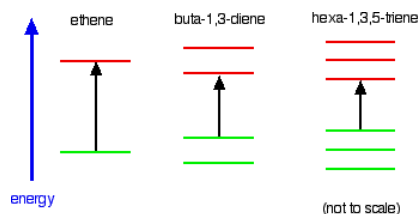
... and that's what is happening.

Compare ethene with buta-1,3-diene. In ethene, there is one pi bonding orbital and one pi anti-bonding orbital. In buta-1,3-diene, there are two pi bonding orbitals and two pi anti-bonding orbitals. This is all discussed in detail on the introductory page that you should have read.



The highest occupied molecular orbital is often referred to as the HOMO - in these cases, it is a pi bonding orbital. The lowest unoccupied molecular orbital (the LUMO) is a pi anti-bonding orbital. Notice that the gap between these has fallen. It takes less energy to excite an electron in the buta-1,3-diene case than with ethene.

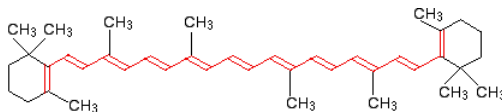
In the hexa-1,3,5-triene case, it is less still.



If you extend this to compounds with really massive delocalisation, the wavelength absorbed will eventually be high enough to be in the visible region of the spectrum, and the compound will then be seen as colored. A good example of this is the orange plant pigment, beta-carotene - present in carrots, for example.

Why is beta-carotene orange?

Beta-carotene has the sort of delocalization that we've just been looking at, but on a much greater scale with 11 carbon-carbon double bonds conjugated together. The diagram shows the structure of beta-carotene with the alternating double and single bonds shown in red.



The more delocalization there is, the smaller the gap between the highest energy pi bonding orbital and the lowest energy pi anti-bonding orbital. To promote an electron therefore takes less energy in beta-carotene than in the cases we've looked at so far - because the gap between the levels is less.

Remember that less energy means a lower frequency of light gets absorbed - and that's equivalent to a longer wavelength. Beta-carotene absorbs throughout the ultra-violet region into the violet - but particularly strongly in the visible region between about 400 and 500 nm with a peak about 470 nm. If you have read the page in this section about electromagnetic radiation, you might remember that the wavelengths associated with the various colors are approximately:

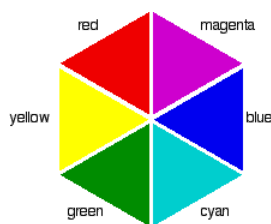
color region	wavelength (nm)
violet	380 - 435
blue	435 - 500
cyan	500 - 520
green	520 - 565
yellow	565 - 590
orange	590 - 625
red	625 - 740

So if the absorption is strongest in the violet to cyan region, what color will you actually see? It is tempting to think that you can work it out from the colors that are left - and in this particular case, you wouldn't be far wrong. Unfortunately, it isn't as simple as that!

Sometimes what you actually see is quite unexpected. Mixing different wavelengths of light doesn't give you the same result as mixing paints or other pigments. You can, however, sometimes get some estimate of the color you would see using the idea of complementary colors.

Complementary colors

If you arrange some colors in a circle, you get a "color wheel". The diagram shows one possible version of this. An internet search will throw up many different versions!



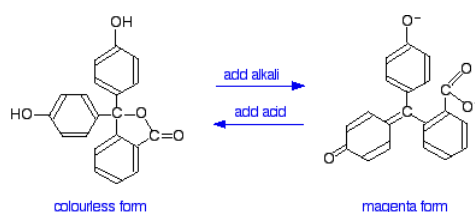
colors directly opposite each other on the color wheel are said to be complementary colors. Blue and yellow are complementary colors; red and cyan are complementary; and so are green and magenta. Mixing together two complementary colors of light will give you white light.

What this all means is that if a particular color is absorbed from white light, what your eye detects by mixing up all the other wavelengths of light is its complementary color. In the beta-carotene case, the situation is more confused because you are absorbing such a range of wavelengths. However, if you think of the peak absorption running from the blue into the cyan, it would be reasonable to think of the color you would see as being opposite that where yellow runs into red - in other words, orange.

Applying this to the color changes of two indicators

Phenolphthalein

You have probably used phenolphthalein as an acid-base indicator, and will know that it is colorless in acidic conditions and magenta (bright pink) in an alkaline solution. How is this color change related to changes in the molecule? The structures of the two differently colored forms are:

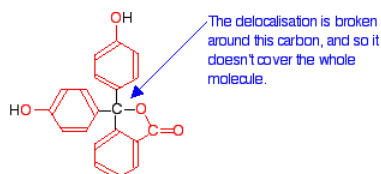


Both of these absorb light in the ultra-violet, but the one on the right also absorbs in the visible with a peak at 553 nm. The molecule in acid solution is colorless because our eyes can't detect the fact that some light is being absorbed in the ultra-violet. However, our eyes do detect the absorption at 553 nm produced by the form in alkaline solution.

553 nm is in the green region of the spectrum. If you look back at the color wheel, you will find that the complementary color of green is magenta - and that's the color you see.

So why does the color change as the structure changes? What we have is a shift to absorption at a higher wavelength in alkaline solution. As we've already seen, a shift to higher wavelength is associated with a greater degree of delocalisation.

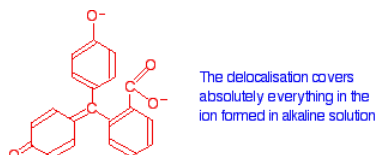
Here is a modified diagram of the structure of the form in acidic solution - the colorless form. The extent of the delocalization is shown in red.



Notice that there is delocalization over each of the three rings - extending out over the carbon-oxygen double bond, and to the various oxygen atoms because of their lone pairs.

But the delocalization doesn't extend over the whole molecule. The carbon atom in the centre with its four single bonds prevents the three delocalized regions interacting with each other.

Now compare that with the magenta form:



The rearrangement now lets the delocalization extend over the entire ion. This greater delocalization lowers the energy gap between the highest occupied molecular orbital and the lowest unoccupied pi anti-bonding orbital. It needs less energy to make the jump and so a longer wavelength of light is absorbed.

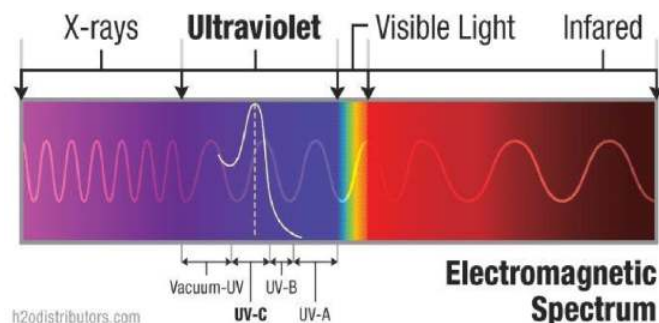
Increasing the amount of delocalization shifts the absorption peak to a higher wavelength.

This page titled [6.1: What Causes Molecules to Absorb UV and Visible Light](#) is shared under a [CC BY-NC 4.0](#) license and was authored, remixed, and/or curated by [Jim Clark](#).

- [What Causes Molecules to Absorb UV and Visible Light](#) by [Jim Clark](#) is licensed [CC BY-NC 4.0](#).

6.2: Solar Energy - UVA, UVB, UVC

Solar energy (sunlight) contains light we can see, and some we cannot. *Visible light* has wavelengths of 750 to 400 nm. *Ultraviolet (UV) light* has shorter wavelengths, cannot be seen, and has higher energy. Infrared (IR) radiation is the major source of heat for Earth. Though UV is a fraction of sunlight, it can be damaging to living organisms. All of these are forms of energy in the *electromagnetic spectrum*.



Reprinted with permission from h2odistributors.com

Just as visible light components have names (red, orange, yellow, green, blue, indigo, violet), so do the types of UV light: UV-A, UV-B, UV-C and vacuum-UV. UV-A has lowest energy and is least damaging; UV-A is also called “black light.” UV-B and UV-C have higher energies and can cause break bonds of molecules, causing changes in DNA and thus skin cancers.

The majority of UV-B is absorbed by ozone in the stratosphere. Though UV-C is most damaging, it is totally absorbed by oxygen and ozone. In recent years, depletion of the ozone layer has allowed more UV light to reach us, resulting in more cases of skin cancers. Consequently, we have become aware of the need to protect ourselves from UV light.

What protects us from UV light? One strategy would be to avoid exposure to any type of sunlight. Since we cannot avoid sunlight while outdoors, we can physically or chemically block the sun. A wide variety of commercial sunscreens are available with *sun protection factors (SPF)* ranging from SPF 2 to SPF 100. These lotions contain organic molecules that absorb UV light. Some materials, such as glass and plastic also absorb UV light, while still allowing visible light through.

UV light type	Wavelength	Relative Energy	Comments
UV-A	320 – 400 nm	lowest energy	reaches Earth in greatest amount
UV-B	280 – 320 nm	higher energy than UV-A, but less than UV-C	most is absorbed by ozone
UV-C	200 – 280 nm	highest energy	absorbed by ozone and oxygen

Questions

1. What is the difference between UV-A, UV-B, and UV-C light?
2. Compare the energies of UV light to IR and visible light. Explain why UV light is potentially more dangerous than IR or visible light.
3. Most sunscreen lotions claim to protect against UV-A and UV-B. Why don't they mention UV-C light?

This page titled [6.2: Solar Energy - UVA, UVB, UVC](#) is shared under a [CC BY-NC](#) license and was authored, remixed, and/or curated by [Santa Monica College](#).

- [4: Detection and Absorption of Ultraviolet Light \(Experiment\)](#) by [Santa Monica College](#) is licensed [CC BY-NC 4.0](#).

6.3: Sunscreen

Exposure to Ultraviolet Radiation

Of all the types of ionizing radiation, people come into more contact with ultraviolet radiation. Sources of this radiation would include the sun, ultraviolet lights, and tanning beds. The sun produces three different forms of ultraviolet rays (UVA, UVB and UVC). Fortunately, the deadliest of these three rays (UVC) never reaches the earth's surface and is absorbed by [Earth's ozone layer](#). However, both UVA and UVB reach the Earth's surface to affect living systems.

UVA is produced naturally by the sun and artificially by lamps and tanning beds. For years, sunbathers believed that UVA provided only a youthful glow. Now, the American Academy of Dermatologists has condemned the use of tanning beds. Scientific research has correlated certain types of skin cancers (mainly basal and squamous cell) with the use of these devices. Besides skin cancer, tanning beds cause premature aging and damage to your eyes. If a tanning bed is not properly sanitized between uses, skin diseases can be transmitted from one consumer to another.

Sunblocks and Sunscreens

UVB radiation in sunlight allows the skin to produce [vitamin D](#). This vitamin prevents bone disorders like rickets and osteoporosis (brittle bone disease). The American Academy of Dermatology suggests vitamin D be obtained through foods or nutritional supplements. Excessive exposure to UV can be damaging and the pigment **melanin**, deposited in cells at the base of the epidermis, helps to protect the underlying layers of the skin from this damage. Melanin also colors the skin and variations in the amount of melanin produces colors from pale yellow to black. The darker the skin tone, the more melanin one has, and the less likely skin cancer will occur.



Figure 6.3.3: The author's great, great- grandmother was diagnosed with skin cancer on her nose during the 1950's.

Excess exposure to the sun can cause sunburn. This is common in humans, but light skinned animals like cats and pigs can also be sunburned, especially on the ears. Skin cancer can also result from excessive exposure to the sun. As holes in the ozone layer increase exposure to the sun's UV rays, so too does the rate of skin cancer in humans and animals.

Sunscreens and sunblocks are designed to protect skin from ultraviolet rays. *Sunblocks* contain inorganic ingredients like zinc oxide or titanium dioxide. These chemicals act as UV filters by reflecting the sun's UV rays. Sunblocks can have grainy textures due to the inorganic components. The thick nature of a sunblock can make it difficult to spread evenly on the skin. *Sunscreens* contain organic compounds like oxybenzone, avobenzone, homosalate, and octinoxate (Figure 6.3.4). By absorbing ultraviolet rays, these compounds decompose and give off heat. Sunscreens apply smoother than sunblocks. Often, manufacturers will combine sunscreen and sunblock ingredients to make their products.

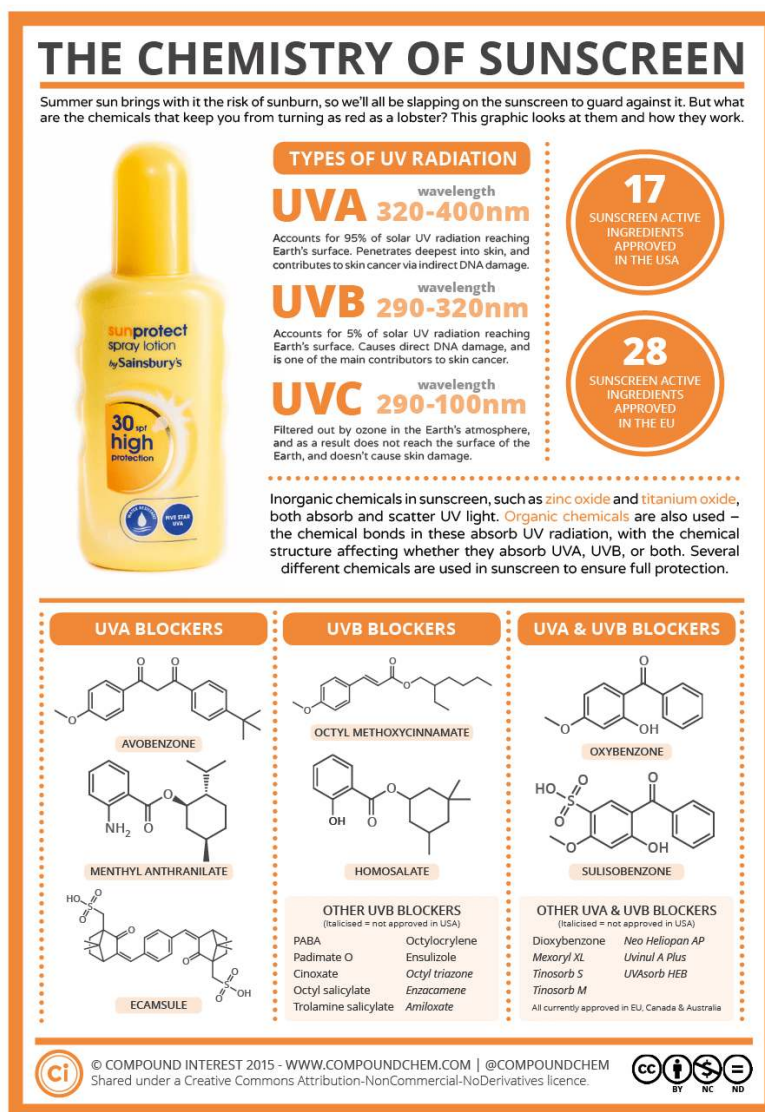


Figure 6.3.4: Effect of Light on skin

Sun protection factor (SPF) measures a product's protection from UVB rays. SPF does not quantify protection from UVA radiation. The American Academy of Dermatology recommends you select a sunscreen or sunblock with a minimum SPF factor of 30. This SPF value means a lotion can filter out 97% of UVB rays. Moving to a SPF of 50 will only filter out 1 more percentage of UVB rays. Increasing a SPF factor past 30 really does very little in shielding skin from UVB radiation.

Products that protect skin from UVA must be labeled as being broad spectrum. Sunblocks provide UVA and UVB protection, but sunscreens can vary on what they can screen. Every two hours, sunscreens/sunblocks should be reapplied to the skin. Sweating and swimming can remove sunscreen/sunblock products. No sun products are waterproof, but some are labeled as being water resistant. Consumers are encouraged to reapply these products every two hours as well.

Contributors

- Elizabeth R. Gordon (Furman University)
- Template:ContribLawson
- Emma Gibney (Furman University)

This page titled 6.3: Sunscreen is shared under a CC BY-NC-SA 4.0 license and was authored, remixed, and/or curated by Elizabeth Gordon.

Index

A

adhesive force

- [3.2: Cohesive and Adhesive Forces](#)
- [3.3: Contact Angles](#)
- [3.4: Capillary Action](#)

adhesive forces

- [3.1: What is surface tension?](#)

B

bomb calorimeter

- [1.3: Constant Volume Calorimetry](#)

C

calorimetry

- [1.1: Calorimetry header](#)
- [1.2: What is Calorimetry?](#)
- [1.3: Constant Volume Calorimetry](#)

capillary action

- [3.4: Capillary Action](#)

Capillary imbibition

- [3.3: Contact Angles](#)

cohesive force

- [3.2: Cohesive and Adhesive Forces](#)
- [3.3: Contact Angles](#)
- [3.4: Capillary Action](#)

cohesive forces

- [3.1: What is surface tension?](#)

Contact Angle

- [3.3: Contact Angles](#)

D

Dynamic Viscosity

- [2.2: Viscosity Theory](#)

F

Fluid Flow

- [2.1: What is Viscosity?](#)

Fluorescence

- [5.1: Fluorescence and Phosphorescence](#)

I

ice calorimeter

- [1.3: Constant Volume Calorimetry](#)

M

M+1 peak (mass spec)

- [4.3: The M+1 Peak](#)

meniscus

- [3.2: Cohesive and Adhesive Forces](#)

molecular ion

- [4.2: Determining the formula of a fragment based on the mass of the Peak](#)

O

Ostwald Viscometer

- [2.2: Viscosity Theory](#)

P

phosphorescence

- [5.1: Fluorescence and Phosphorescence](#)

S

Sessile Drop Method

- [3.3: Contact Angles](#)

surface tension

- [3.1: What is surface tension?](#)
- [3.3: Contact Angles](#)

T

Tears of Wine

- [3.2: Cohesive and Adhesive Forces](#)

triplet excited state

- [5.1: Fluorescence and Phosphorescence](#)

V

viscosity

- [2.1: What is Viscosity?](#)
- [2.2: Viscosity Theory](#)

W

wetting agent

- [3.2: Cohesive and Adhesive Forces](#)

Wilhelmy Plate Method

- [3.3: Contact Angles](#)

Y

Young's equation

- [3.3: Contact Angles](#)

Detailed Licensing

Overview

Title: [MSU: CHMY 362 Elements of Physical Chemistry](#)

Webpages: 32

Applicable Restrictions: Noncommercial

All licenses found:

- [Undeclared](#): 65.6% (21 pages)
- [CC BY-NC 4.0](#): 12.5% (4 pages)
- [CC BY 4.0](#): 9.4% (3 pages)
- [CC BY-NC-SA 4.0](#): 6.3% (2 pages)
- [CC BY-SA 4.0](#): 3.1% (1 page)
- [CC BY 3.0](#): 3.1% (1 page)

By Page

- [MSU: CHMY 362 Elements of Physical Chemistry - Undeclared](#)
 - [Front Matter - Undeclared](#)
 - [TitlePage - Undeclared](#)
 - [InfoPage - Undeclared](#)
 - [Table of Contents - Undeclared](#)
 - [Licensing - Undeclared](#)
 - [1: Calorimetry - Undeclared](#)
 - [1.1: Calorimetry header - Undeclared](#)
 - [1.2: What is Calorimetry? - CC BY-SA 4.0](#)
 - [1.3: Constant Volume Calorimetry - CC BY 3.0](#)
 - [2: Viscosity - Undeclared](#)
 - [2.1: What is Viscosity? - CC BY-NC-SA 4.0](#)
 - [2.2: Viscosity Theory - CC BY 4.0](#)
 - [3: Surface Tension - Undeclared](#)
 - [3.1: What is surface tension? - Undeclared](#)
 - [3.2: Cohesive and Adhesive Forces - CC BY 4.0](#)
 - [3.3: Contact Angles - Undeclared](#)
 - [3.4: Capillary Action - Undeclared](#)
 - [4: MALDI-TOF - Undeclared](#)
 - [4.1: Overview of mass spectrometry - Undeclared](#)
 - [4.2: Determining the formula of a fragment based on the mass of the Peak - CC BY-NC 4.0](#)
 - [4.3: The M+1 Peak - CC BY-NC 4.0](#)
 - [5: Fluorometry - Undeclared](#)
 - [5.1: Fluorescence and Phosphorescence - CC BY 4.0](#)
 - [6: UV-Vis - Undeclared](#)
 - [6.1: What Causes Molecules to Absorb UV and Visible Light - CC BY-NC 4.0](#)
 - [6.2: Solar Energy - UVA, UVB, UVC - CC BY-NC 4.0](#)
 - [6.3: Sunscreen - CC BY-NC-SA 4.0](#)
 - [Back Matter - Undeclared](#)
 - [Index - Undeclared](#)
 - [Glossary - Undeclared](#)
 - [Detailed Licensing - Undeclared](#)